

Modelling Financial Markets

Chapter 9: Realised Volatility



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From Intraday Prices to Variance Forecasts

What we will explain

- Part I, **measurement and diagnosis**: realised variance and its precision, microstructure noise, jumps
- Part II, **modelling and forecasting**
  - stylised facts, the heterogeneous autoregressive (HAR) model, forecast evaluation against GARCH (generalised autoregressive conditional heteroskedasticity, Chapter 5)
  - rough volatility, a machine-learning case study, the research loop assisted by artificial intelligence (AI): extensions that build on the core path, which ends with the model confidence set
- Asymptotic theory, noise- and jump-robust estimators and model extensions sit in the Appendix at the end

What you should be able to do

- compute daily realised variance, its standard error and a confidence interval from intraday prices
- diagnose microstructure noise with a signature plot and test for jumps with false discovery rate (FDR) control
- estimate log-HAR and compare variance forecasts with the quasi-likelihood loss (QLIKE), Diebold–Mariano, Mincer–Zarnowitz and the model confidence set

Volatility Is Latent: Can We Measure It?



Torben G. Andersen, 2012

Photo: Soralum (2012); CC BY-SA 4.0; Wikimedia Commons

- Volatility drives risk (Chapters 7–8), option prices (Chapter 12) and portfolio weights (Chapter 4), but it is never observed
  - One daily return is a single draw from a distribution whose dispersion we want to estimate
  - GARCH (Chapter 5) **infers** today's variance from past daily returns through a model
- Idea: inside one day there are hundreds of price changes; the sum of their squares **measures** the day's variance almost without a model Andersen & Bollerslev (1998)
  - Early idea: Merton (1980) showed that variance, unlike the mean, is estimated better with more frequent data
- Two different objects
  - realised variance: an **ex-post measurement** of day t, known after the close
  - GARCH variance: a **conditional forecast** of day t made with the information of day $t-1$
- Measure first, then model
  - Andersen, Bollerslev, Diebold & Labys (2001), Andersen, Bollerslev, Diebold & Ebens (2001) study the distribution of realised volatility
  - Andersen, Bollerslev, Diebold & Labys (2003) forecast it directly; textbook treatment: Franke, Härdle & Hafner (2019)

Reading Intraday Data

- SPY tracks the Standard & Poor's 500 stock index through an Exchange-Traded Fund (ETF), a fund whose units trade on an exchange
- UTC means Coordinated Universal Time: Bitcoin daily intervals run midnight to midnight in that time zone
- VIX is the Volatility Index of the Chicago Board Options Exchange, an option-price measure of expected volatility
- Up to 18 September 2026; returns $r_i = 100(\ln P_i - \ln P_{i-1})$ in %, P_i: the price at the end of bar i; a bar is labelled by its start time
- Charts and times quoted in this chapter mark the **end** of each 5-minute interval: 09:35 is the first SPY return, 09:30–09:35
- Log returns, not simple percentage changes $100(P_i - P_{i-1})/P_{i-1}$: log returns add up over bars, so the daily return is the sum of the intraday ones; SPY overnight on 7 April 2025: log return -3.27%, simple change -3.22%

Two Markets, Two Clocks: SPY and Bitcoin

Series	Frequency	Period	Days
SPY (ETF)	5-minute bars, 09:30–16:00 New York time	2020–2026	1 489
Bitcoin	5-minute bars, 24-hour days (UTC)	2024–2026	659
SPY, Bitcoin	daily closes	2000/2014–2026	for GARCH
VIX	daily close	2020–2026	implied volatility

- SPY: 78 returns per full day, 43 on the 12 half-days; bars off the 5-minute grid removed; missing bars inside a day carry the last price
- Bitcoin: 288 returns per day; days with fewer than 95% of the bars dropped (88 of 747 calendar days, mostly a data gap in October 2025)

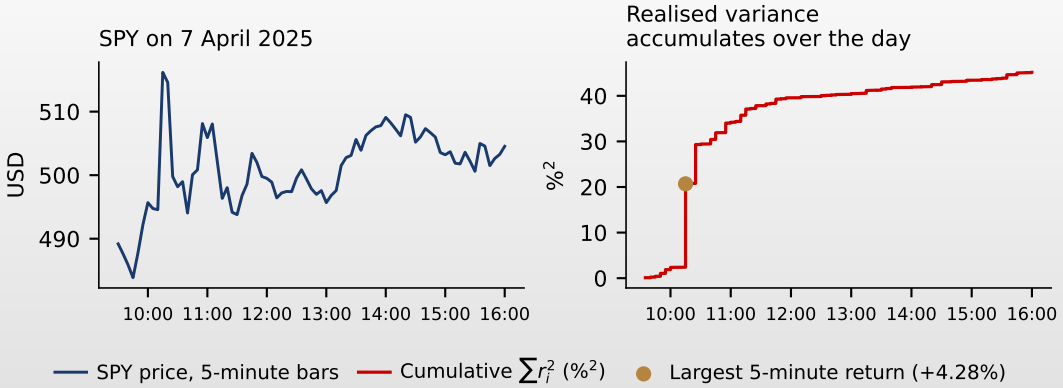
A Model for Intraday Prices

- Notation: day t has M intervals of length $\Delta = 1/M$, returns $r_{t,i}$, daily return $r_t = \sum_i r_{t,i}$
  - Variances in $\%^2$; annualised volatility $\sqrt{252 \times \text{variance}}$ (SPY), $\sqrt{365 \times \text{variance}}$ (Bitcoin); the SPY overnight return (close to next open) is outside the bars
- Log price in continuous time: $dp(s) = \mu(s)\,ds + \sigma(s)\,dW(s)$, with W a Brownian motion (Chapter 11)
  - In words: over an instant ds the log price moves by a drift $\mu(s)\,ds$ (the instantaneous mean return) plus a random shock $\sigma(s)\,dW(s)$, with $dW(s) \sim N(0, ds)$
  - $\sigma(s)$ is the **spot volatility**: it changes within the day and from day to day
- **Integrated variance** of day t: the quantity we want to measure
  - $\mathrm{IV}_t = \int_{t-1}^{t} \sigma^2(s)\,ds$: the spot variance accumulated over the day; in this chapter IV always means integrated variance, never implied volatility
- If σ is independent of W (no leverage) and the drift is zero (or fixed by the path of σ), then given that path the daily return is Normal: $r_t \mid \sigma \sim N(\int \mu, \mathrm{IV}_t)$
  - With leverage (Appendix: Itô semimartingales) this conditional normality fails, but the limit theory of realised variance (RV, next slide) still holds

Definition: Realised Variance

- **Realised variance**: the sum of squared intraday returns
  - $\mathrm{RV}_t = \sum_{i=1}^{M} r_{t,i}^2$, realised volatility $= \sqrt{\mathrm{RV}_t}$
- As $M \to \infty$, $\mathrm{RV}_t \to \mathrm{IV}_t$: the drift vanishes and the sum of squares converges to the **quadratic variation** Barndorff-Nielsen & Shephard (2002)
  - The mean of SPY 5-minute returns is below 1% of their standard deviation: it can be ignored
- No parameters to estimate: RV_t is a **model-free** measurement of one day's variance
- The squared daily return r_t^2 is the special case $M = 1$: unbiased (with zero drift) but extremely noisy

One Day, 78 Returns: 7 April 2025 (1/2)



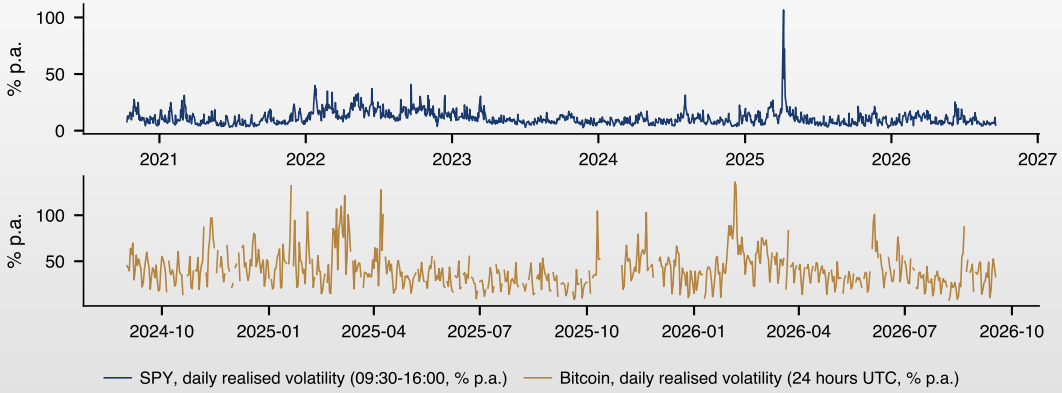
MFM_ch9_intraday_day

One Day, 78 Returns: 7 April 2025 (2/2)

- Left: SPY on 5-minute bars on the most volatile day of the sample; right: running sum of squared returns
- SPY opened with an overnight log return of -3.27%, after tariff announcements; 78 five-minute returns followed
  - $\mathrm{RV} = 45.1\%^2$: annualised realised volatility $\sqrt{252 \times 45.1} = 107\%$
- One 5-minute return of $+4.28\%$ (interval ending 10:15) contributed 41% of the day's realised variance
- Open-to-close the day ended $+3.09\%$: the daily return hides the intraday swings that RV records

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Daily Realised Volatility: SPY and Bitcoin (1/2)



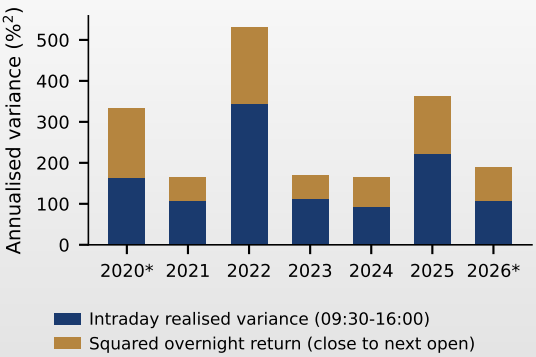
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Daily Realised Volatility: SPY and Bitcoin (2/2)

- Annualised, from 5-minute returns; SPY: regular US session; Bitcoin: 24 hours; the gap in October 2025 is missing data
- SPY: volatility from mean RV $\sqrt{252\,\overline{\mathrm{RV}}} = 12.9\%$ p.a., median day 9.7%, range 2.4% to 106.7%
  - Volatility computed from mean variance exceeds the median: a few very volatile days raise the root-mean-square of daily volatilities
- Bitcoin: $\sqrt{365\,\overline{\mathrm{RV}}} = 45.1\%$ p.a., median 37.8%, maximum 136.1% on 2026-02-05
- Both series show **clustering**, as with GARCH (Chapter 5), but every point is a measurement, not a model output

MFM_ch9_rv_series

Where Is the Risk: Day or Night?



- SPY, mean daily variance by year, annualised: intraday realised variance plus squared overnight return; 2020 and 2026 are partial years
- Overnight returns carry 38% of SPY's daily variance, although the market is closed for 17.5 hours
  - News (earnings, macro data, Asian and European sessions) accumulates and is priced at the open
- Annualised: close-to-close 16.6%; open-to-close 13.6%; overnight 10.1%
- Largest overnight move: -4.08% on 2024-08-05
- For daily risk we use **total** variance: $\mathrm{RV}_t + r^2_{\mathbf{overnight},t}$ (annualised mean 16.4%)
  - Alternatives: scale intraday RV up, or weight the two parts optimally Hansen & Lunde (2005)

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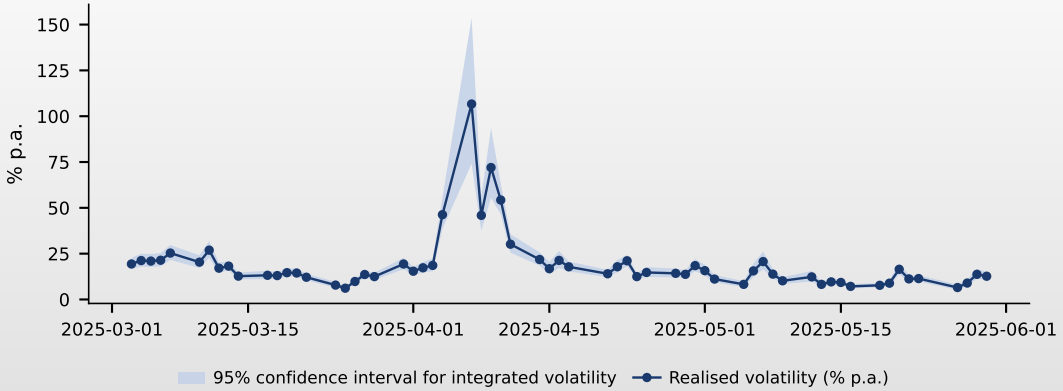
Adding Up Day and Night Variance

- SPY, annualised variances (%^{2} per year)
  - intraday RV: 12.9% (slide *Daily Realised Volatility*), variance 167
  - total variance $\mathrm{RV}_t + r^2_{\text{overnight},t}$: 16.4%, variance 270
  - overnight part: $270 - 167 = 103$, i.e. $\sqrt{103} \approx 10.1\%$; share $103/270 = 38\%$
- Why not $13.6^2 + 10.1^2$? The open-to-close 13.6% squares the whole session return, not each 5-minute return
  - $(\sum_i r_{t,i})^2 = \mathrm{RV}_t + 2\sum_{i<j} r_{t,i}\, r_{t,j}$: the cross products are zero only in expectation; in this sample they add about 17 ($13.6^2 \approx 184$)
  - close-to-close (16.6%) also contains the cross term $2\, r_{\text{overnight},t}\, r_{\text{open-close},t}$

How Precise Is Realised Variance?

- Feasible central limit theorem (CLT) for RV Barndorff-Nielsen & Shephard (2002): $\sqrt{M}\,(\mathrm{RV}_t - \mathrm{IV}_t)/\sqrt{2\,\mathrm{RQ}_t} \to N(0,1)$
  - In words: the error $\mathrm{RV}_t - \mathrm{IV}_t$, scaled by $\sqrt{M}$ and by its estimated standard deviation, is approximately standard Normal
  - $\mathrm{IQ}_t = \int_{t-1}^{t} \sigma^4(s)\,ds$: the integrated quarticity, which sets the variance of RV; it is estimated by the realised quarticity $\mathrm{RQ}_t = \frac{M}{3}\sum_i r_{t,i}^4$
- Standard error: $\mathrm{se}(\mathrm{RV}_t) = \sqrt{\frac{2}{3}\sum_i r_{t,i}^4}$
- 95% confidence interval (CI) on the log scale (always positive): $\mathrm{RV}_t \exp\left(\pm 1.96\,\mathrm{se}(\mathrm{RV}_t)/\mathrm{RV}_t\right)$
  - With constant volatility, $\mathrm{se}/\mathrm{RV} \approx \sqrt{2/M}$: 16% for $M = 78$
- The limiting variance $2\,\mathrm{IQ}_t$ is itself random (a mixed Normal limit); replacing it by $2\,\mathrm{RQ}_t$ is legitimate because the convergence is **stable** (formal statement in the Appendix)

Confidence Intervals for Daily Volatility (1/2)



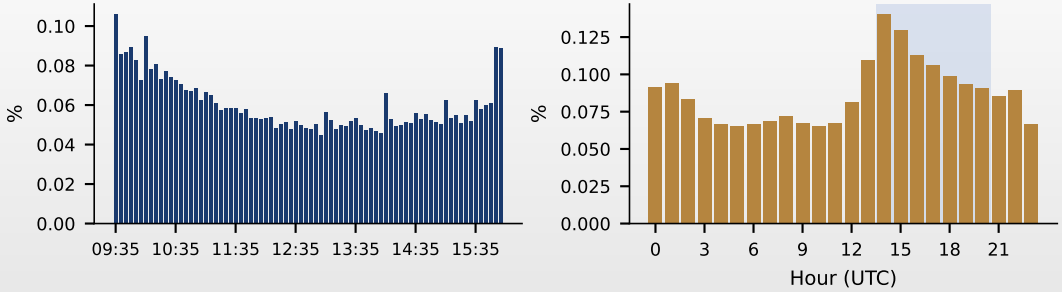
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Confidence Intervals for Daily Volatility (2/2)

- SPY, March–May 2025; band: 95% confidence interval for $\sqrt{252\,\mathrm{IV}_t}$ from realised quarticity
- Median relative width of the interval (for volatility): 37% of its level
  - The band assumes a continuous price and negligible noise: on a jump day RV tends to $\mathrm{IV}_t + \sum_j \kappa_j^2$, and $\sum r^4$ keeps the term κ^4, so the band has no nominal coverage for either the total variation or IV_t (section Jumps)
- 9 April 2025 (the day the tariff pause was announced): volatility 72%, CI [55%, 93%]
  - The interval widens when a few returns dominate the sum: $\sum r^4$ grows faster than $(\sum r^2)^2$
- From variance to volatility: the band takes the square root of the variance endpoints, $\sqrt{\mathrm{RV}_t}\exp\left(\pm 0.98\,\mathrm{se}(\mathrm{RV}_t)/\mathrm{RV}_t\right)$, so the log half-width is halved
  - SPY: median $\mathrm{se}/\mathrm{RV} = 0.19$, so the volatility factors are $[0.83, 1.20]$: the median width of 37%
- Still far more precise than r_t^2, whose relative error is of order $\sqrt{2} \approx 141\%$ under the Normal distribution
- Practical rule: a single day's volatility is known to about $\pm 20\%$ (95% band), averages over weeks much more precisely

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Intraday Seasonality (1/2)



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Intraday Seasonality (2/2)

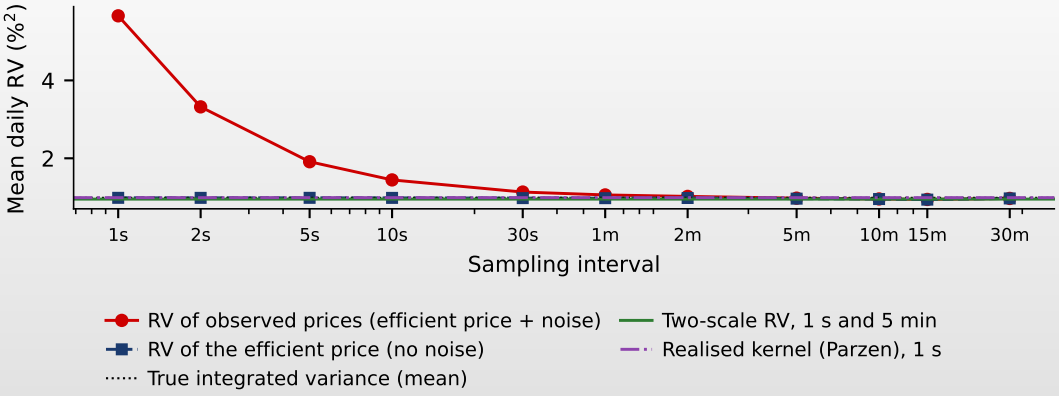
- Left: SPY, mean $|r_{t,i}|$ by 5-minute interval of the day (full days only; ticks mark the end of the interval, New York time); right: Bitcoin by UTC hour
- SPY: a **U-shape**: 0.106% in the first 5 minutes, 0.050% around midday, 0.089% in the last 5 minutes Andersen & Bollerslev (1997)
  - The opening auction incorporates overnight news; the close concentrates index and fund rebalancing
- Bitcoin trades 24/7 but is most volatile at 14:00 UTC, near the US equity opening (09:30 New York: 13:30 UTC in summer, 14:30 UTC in winter); quietest at 10:00 UTC (ratio 2.2)
- RV sums over the day, so seasonality does not bias it
  - it does bias bipower variation (BV, defined in section Jumps) and makes the jump test over-reject
  - the remedy, dividing returns by a periodicity factor f_i Boudt, Croux & Laurent (2011), is applied to the jump test in section Jumps (▸ Appendix)

MFM_ch9_seasonality

Why Not Use Every Tick?

- Observed prices are not the efficient price: $p_i^{\text{obs}} = p_i^* + \varepsilon_i$, so $r_i = r_i^* + \varepsilon_i - \varepsilon_{i-1}$
  - p_i^*: the efficient (true) log price, ε_i: the noise in the observed price, r_i^*: the efficient return
  - Sources of ε: bid–ask bounce, tick size (price discreteness), stale quotes (Chapter 10)
- With i.i.d. noise of variance ω^2 and n returns per day Zhang, Mykland & Aït-Sahalia (2005), Hansen & Lunde (2006)
  - $\mathbb{E}[\mathrm{RV}_t^{(n)} \mid \sigma] = \mathrm{IV}_t + 2n\omega^2$ (zero drift, no leverage): each squared return carries $2\omega^2$ of noise, so the bias **grows** with the sampling frequency
  - Adjacent returns share ε_{i-1} with opposite signs: $\mathrm{cov}(r_i, r_{i-1}) = -\omega^2$
- Trade-off: $\mathbb{E}[(\mathrm{RV}^{(n)} - \mathrm{IV})^2] \approx 2\,\mathrm{IQ}/n + 4n^2\omega^4$, discretisation error plus squared bias
  - MSE (mean squared error) optimal number of returns $n^* = (\mathrm{IQ}/(4\omega^4))^{1/3}$ Bandi & Russell (2008): more returns cut the first term, but raise the second; derivation in the ▸ Appendix
- Numbers: $\mathrm{IV} = 1\%^2$ with constant volatility within the day, so $\mathrm{IQ} = \mathrm{IV}^2 = 1\ (\%^4)$; $\omega = 0.01\%$, so $\omega^2 = 10^{-4}\%^2$: bias 0.016 at 5 minutes ($n = 78$), 4.680 at 1 second; $n^* = 292$ (one return every 80 s)

The Volatility Signature Plot: Simulation (1/2)



MFM_ch9_noise_simulation

The Volatility Signature Plot: Simulation (2/2)

- A signature plot shows mean RV against the sampling interval; here: 100 simulated days, one price per second; efficient price with random daily volatility plus i.i.d. noise, $\omega = 0.01\%$
- True mean IV $= 0.98$; RV of observed prices: 5.66 at 1 second, 1.06 at 1 minute, 0.98 at 5 minutes
  - The bias matches $2n\omega^2$ (slide *Why Not Use Every Tick?*)
- Without noise, RV is flat in the sampling interval: the rise at high frequency is the fingerprint of noise
- Noise-robust estimators on 1-second data: two-scale RV 0.95, realised kernel 0.99
  - error standard deviation across days: 5-minute RV 0.187, two-scale 0.150, kernel 0.045
  - using all data pays off only with a noise correction

MFM_ch9_noise_simulation

Remedies: Sparse Sampling and Subsampling

- **Sparse sampling**: use one return every 5 minutes instead of every tick
  - Simple and robust; throws away most of the data Liu, Patton & Sheppard (2015)
- **Subsampling**: compute RV on K shifted grids (offsets $0, 1, \ldots, K-1$) and average; $\mathrm{RV}_t^{(k)}$: RV on the grid with offset k
  - $\mathrm{RV}_t^{\mathsf{avg}} = \frac{1}{K}\sum_{k=0}^{K-1} \mathrm{RV}_t^{(k)}$: same bias as one sparse grid, smaller variance
- **Two-scale RV** (TSRV) Zhang, Mykland & Aït-Sahalia (2005): remove the remaining bias with the all-data RV
  - $\mathrm{TSRV}_t = \mathrm{RV}_t^{\mathsf{avg}} - \frac{\bar{n}}{n}\mathrm{RV}_t^{\mathsf{all}}$, $\bar{n} = (n - K + 1)/K$; $\mathrm{RV}_t^{\mathsf{all}}$: RV on all n returns, which estimates the noise term $2n\omega^2$; divide by $1 - \bar{n}/n$ in small samples
  - $\bar{n}$ is the average number of returns per sparse grid; every grid includes the first and the last price of the day

Realised Kernels

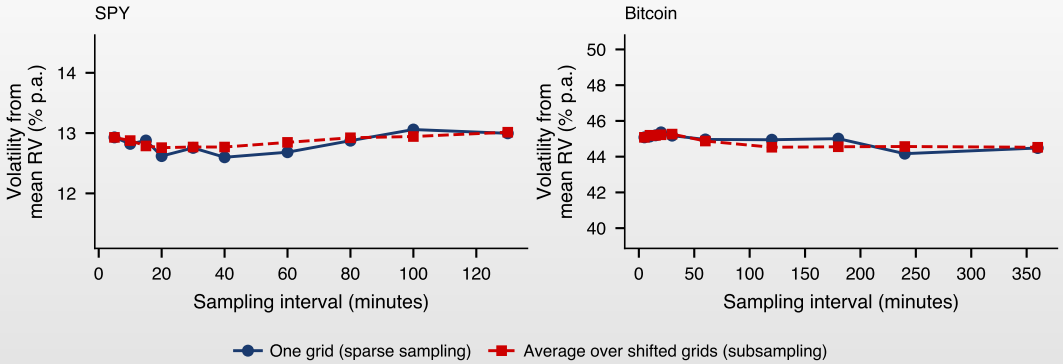


Neil Shephard, 2004

Photo: Renate Schmid (2004); CC BY-SA 2.0 de; Wikimedia Commons

- **Realised kernel** Barndorff-Nielsen, Hansen, Lunde & Shephard (2008): add weighted autocovariances of returns
  - $\mathrm{RK}_t = \gamma_0 + 2\sum_{h=1}^{H} k\left(\frac{h}{H+1}\right)\gamma_h$, $\gamma_h = \sum_i r_{t,i}\, r_{t,i-h}$
  - Parzen weight function k: 1 at 0, falling smoothly to 0 at 1; keeps RK positive
- Noise makes γ_1 negative; the kernel adds it back and cancels the bias
- Bandwidth H grows with the noise-to-signal ratio
  - $H \approx 3.5\,\xi^{4/5} n^{3/5}$, with $\xi^2 = \omega^2/\sqrt{\mathrm{IQ}}$ the noise-to-signal ratio
- On liquid 5-minute data the kernel changes RV only slightly (next slides); it pays off with tick data

The Signature Plot on Real Data (1/2)



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The Signature Plot on Real Data (2/2)

- Annualised volatility from mean RV by sampling interval: one grid vs the average over all shifted grids
- Flat: SPY 12.9% at 5 minutes, 12.8% at 30, 13.0% at 130; Bitcoin 45.1% at 5 minutes, 44.5% at 6 hours
  - First-order autocorrelation of 5-minute returns: -0.008 (SPY), 0.005 (Bitcoin): no visible noise at 5 minutes
- Limit of this dataset: 5-minute bars can diagnose 5-minute and coarser sampling only; they say nothing about noise below 5 minutes
- On 5-minute SPY data: two-scale RV / RV $= 0.97$, realised kernel / RV $= 0.95$; correlations with RV 0.89 and 0.85
- Consistent with Liu, Patton & Sheppard (2015): almost 400 estimators on 31 assets rarely beat 5-minute RV
  - noise-robust estimators pay off when noise is large relative to the signal (tick or second-level data, illiquid assets)

MFM_ch9_signature

Prices Jump: Bipower Variation



Ole E. Barndorff-Nielsen, 2007

Photo: Thomas Steiner (2007); CC BY-SA 2.5; Wikimedia Commons

- Add jumps to the model: $dp = \mu\, ds + \sigma\, dW + dJ$, with J a jump process (news, Chapter 11)
  - Quadratic variation now has two parts: $\mathrm{RV}_t \to \mathrm{IV}_t + \sum_j \kappa_j^2$, with κ_j the jump sizes in day t
  - Why separate them? Jumps are less persistent than diffusive volatility, and they price differently (options, Chapter 12)
- **Bipower variation** Barndorff-Nielsen & Shephard (2004): products of adjacent absolute returns
  - $\mathrm{BV}_t = \mu_1^{-2} \frac{M}{M-1} \sum_{i=2}^{M} |r_{t,i}|\,|r_{t,i-1}| \to \mathrm{IV}_t$
  - $\mu_1 = \mathbb{E}|Z| = \sqrt{2/\pi} = 0.798$ for $Z \sim N(0,1)$, so $\mu_1^{-2} = \pi/2 = 1.571$; $\frac{M}{M-1}$: finite-sample normalisation
- A jump enters one return, which is multiplied by a small neighbour: its effect on BV vanishes as M grows; the jump part of the day is $\max(\mathrm{RV}_t - \mathrm{BV}_t, 0)$
  - Example, $M = 4$ returns $0.1, 0.1, 0.1, 1.0$ (%): $\mathrm{RV} = 1.03$, $\mathrm{BV} = \frac{\pi}{2} \cdot \frac{4}{3}(0.01 + 0.01 + 0.10) = 0.25$, jump part 0.78 (%^{2})

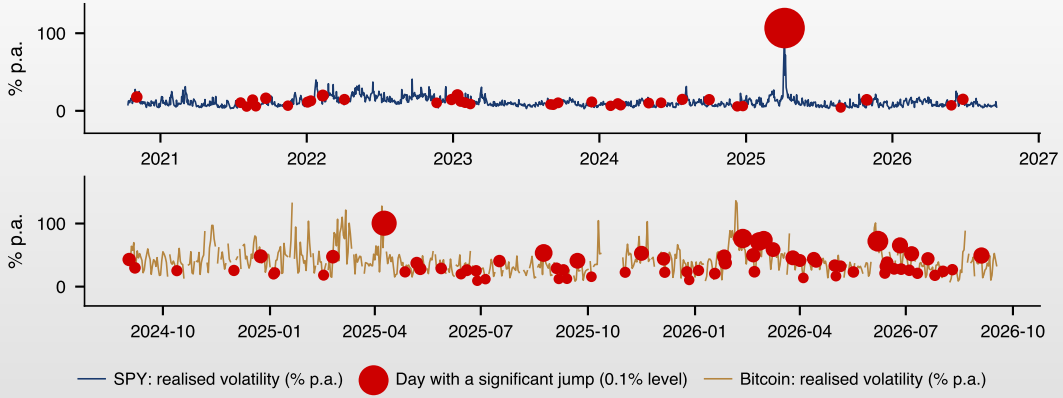
A Test for Jumps

- Ratio statistic Huang & Tauchen (2005), Barndorff-Nielsen & Shephard (2006):

$$z_t = \frac{(\mathrm{RV}_t - \mathrm{BV}_t)/\mathrm{RV}_t}{\sqrt{\theta\, M^{-1} \max\left(1, \mathrm{TQ}_t/\mathrm{BV}_t^2\right)}}$$

  - $\theta = \pi^2/4 + \pi - 5 = 0.609$; tripower quarticity $\mathrm{TQ}_t = M\mu_{4/3}^{-3} \frac{M}{M-2} \sum_i |r_{t,i}|^{4/3} |r_{t,i-1}|^{4/3} |r_{t,i-2}|^{4/3}$, $\mu_{4/3} = \mathbb{E}|Z|^{4/3}$
  - In words: the relative jump share $(\mathrm{RV}_t - \mathrm{BV}_t)/\mathrm{RV}_t$ divided by its standard error; θ: the asymptotic variance constant; TQ estimates IQ_t robustly to jumps
- Under no jumps $z_t \to N(0,1)$; a jump makes $\mathrm{RV} > \mathrm{BV}$ and z_t large and positive
- One test per day: at level α we expect $\alpha\times$ (number of days without a jump) false alarms, at most $\alpha\times$ days
  - We use $\alpha = 0.1\%$ (critical value 3.09) for the jump days plotted on the next slide
- Significant day: jump part $J_t = \mathrm{RV}_t - \mathrm{BV}_t$, continuous part $C_t = \mathrm{RV}_t - J_t$; otherwise $J_t = 0$

Jump Days: SPY and Bitcoin (1/2)



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Jump Days: SPY and Bitcoin (2/2)

- Red points: days with a significant jump at the 0.1% level; point size proportional to the jump part J_t
- SPY: 35 jump days of 1 489 (2.4%); jumps are 2.8% of total realised variance
  - on a jump day the jump is on average 35% of that day's RV
  - BV/RV averages 0.95 over all days
  - largest statistic: 2021-08-27 ($z = 5.2$), 10:00–10:05: the Jackson Hole speech of the Fed chair
- Bitcoin: 67 jump days of 659 (10.2%), but only 1.8% of variance
  - Many small jumps on quiet days (weekends): one isolated move dominates a calm day
- The turbulent days of April 2025 are mostly **not** jump days: many large returns in a row raise BV as well

MFM_ch9_jumps

Many Days, Many Tests: Spurious Jumps

- 1 489 daily tests at 5%: 235 rejections against 74 expected under no jumps; which of them are false?
- Part of the excess is intraday seasonality (section Realised Variance): dividing returns by the periodicity factor f_i Boudt, Croux & Laurent (2011) already cuts the 5% rejections from 235 to 167
- Benjamini & Hochberg (1995): sort the p-values and reject the k smallest
  - k: the largest index with $p_{(k)} \leq kq/N$; $p_{(k)}$: k-th smallest p-value, N: number of tests, q: target FDR
  - here each day's p-value is the one-sided $1 - \Phi(z_t)$ of the ratio statistic
  - the FDR, $\mathbb{E}$[false/rejections], stays $\leq q$ (false/rejections $:= 0$ when nothing is rejected)
  - The guarantee needs valid p-values that are independent or positively dependent Benjamini & Yekutieli (2001); daily jump tests are only approximately so

The Benjamini–Hochberg Rule on Five Tests

- Five days, sorted p-values $0.003, 0.012, 0.024, 0.045, 0.60$; target $q = 0.05$, $N = 5$
- Thresholds kq/N for $k = 1, \ldots, 5$: $0.01, 0.02, 0.03, 0.04, 0.05$
- Compare position by position
  - $0.003 \leq 0.01$, $0.012 \leq 0.02$, $0.024 \leq 0.03$: pass; $0.045 > 0.04$, $0.60 > 0.05$: fail
  - largest passing index $k = 3$: reject the three smallest p-values
- Three rules, three answers on the same data
  - no correction ($p \leq 0.05$): 4 rejections; BH at $q = 5\%$: 3; Bonferroni ($p \leq 0.05/5 = 0.01$): 1
- Check (step-up): with p-values $0.003, 0.012, 0.032, 0.038, 0.60$, the third fails its own threshold ($0.032 > 0.03$) but the fourth passes ($0.038 \leq 0.04$), so $k = 4$ and four days are rejected

Jump Counts under Multiplicity Control

- SPY, 1 489 days
  - FDR at 5%: 42 jump days; at 1%: 9; Bonferroni (family-wise error rate, FWER, 5%): 7; FDR 5% after the periodicity correction: 17
  - from 235 unadjusted rejections to 17: most rejections do not survive the two corrections for seasonality and multiplicity
- Bajgrowicz, Scaillet & Treccani (2016): without multiplicity control most detected jumps are spurious; the true ones are rare and cluster on news
- Christensen, Oomen & Podolskij (2014): at tick frequency most 5-minute "jumps" are steep but continuous bursts of volatility (fact or friction)
- A daily test answers "did day t contain a jump?"; locating the jump within the day needs an intraday test (Lee–Mykland, ▸ Appendix)

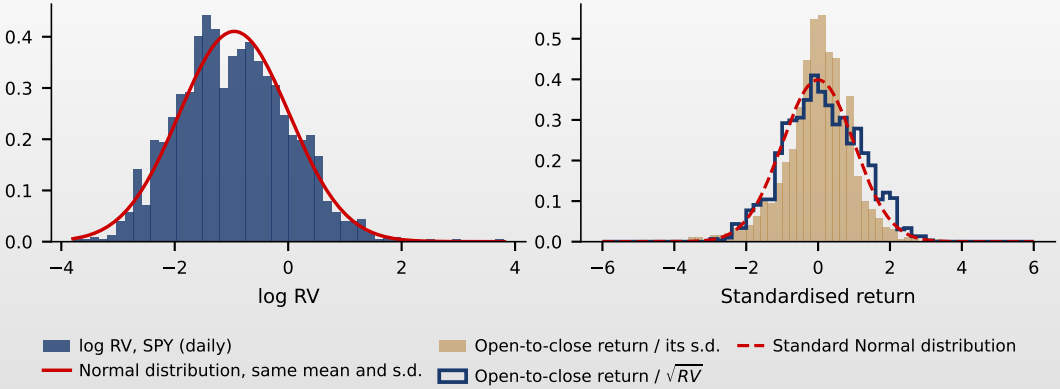
Check Yourself: Measuring Daily Variance

- Why does RV converge to integrated variance as M grows?
  - The sum of squared returns converges to the quadratic variation, and the drift contribution, of order Δ^2 per interval, vanishes
- Why does the standard error of RV use fourth powers of returns?
  - The variance of RV depends on integrated quarticity IQ_t, estimated by $\mathrm{RQ}_t = \frac{M}{3} \sum_i r_{t,i}^4$
- Why is a 95% band for volatility narrower, on the log scale, than the band for variance?
  - The square root halves the log half-width: the volatility factors are the square roots of the variance factors
- Why is SPY's daily risk understated if you use intraday RV alone?
  - Overnight returns bring 38% of SPY daily variance

Check Yourself: Noise and Jumps

- Why does sampling every tick overstate variance?
  - Every squared return adds $2\omega^2$ of noise, so the bias $2n\omega^2$ grows with the number of returns
- What does a flat signature plot tell you? What does it not tell you?
  - It shows that noise is negligible at 5 minutes and coarser; it says nothing about noise below 5 minutes
- Why does bipower variation stay close to IV on a jump day while RV does not?
  - The jump enters a single return: squared in RV, multiplied by a small neighbour in BV
- With 1 489 daily jump tests, why is "$p < 0.05$" not evidence of a jump? What does the FDR control?
  - At 5%, 1 489 jump-free days would give about 74 false rejections; the FDR controls the expected share of false rejections among rejections

Distributions: log RV and Standardised Returns (1/2)



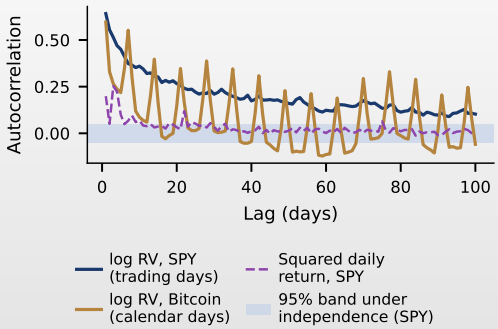
MFM_ch9_distributions

Distributions: log RV and Standardised Returns (2/2)

- Left: daily log RV of SPY and the Normal density; right: open-to-close returns scaled by their standard deviation (filled) and by $\sqrt{\mathrm{RV}_t}$ (outline)
- SPY RV: skewness 20.0, kurtosis 550; log RV: skewness 0.29, kurtosis 3.20 (the Normal distribution: 0 and 3)
  - Bitcoin log RV: skewness -0.27, kurtosis 3.33: model volatility in logs
- Open-to-close returns: kurtosis 15.73; divided by $\sqrt{\mathrm{RV}_t}$: 2.65, standard deviation 1.05
  - Bitcoin: from 7.57 to 2.77
- Heavy tails of daily returns (Chapter 1) come mostly, ex post, from changing volatility: approximately a **mixture** of Normal distributions with variances IV_t, exact only without leverage and jumps (section Realised Variance)

MFM_ch9_distributions

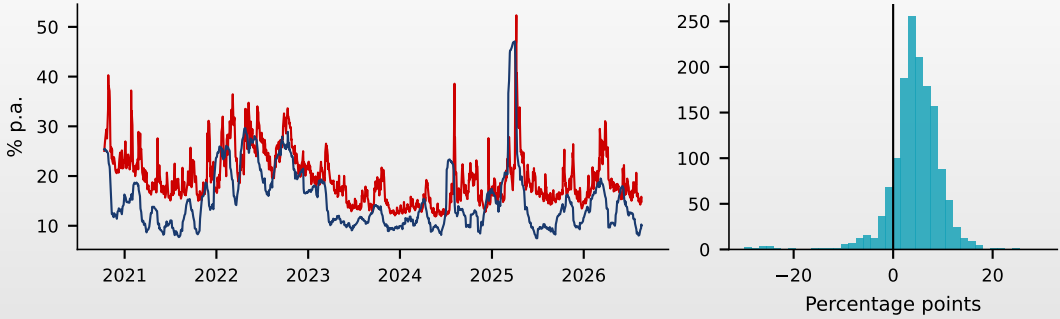
Persistence of Realised Volatility



- Autocorrelations up to 100 days: log RV of SPY (total variance) and Bitcoin, squared daily SPY returns
- SPY log RV: 0.64 at lag 1, 0.24 at one month, still 0.10 at 100 days: **highly persistent** dependence
  - an autoregressive AR(1) model would decay geometrically: $0.64^{22} \approx 0.00005$
  - a sample autocorrelation function (ACF) is consistent with long memory but cannot prove it
- Squared daily returns show the same persistence, hidden by noise: 0.20 at lag 1
- Bitcoin: peaks at 7, 14, 21, ... days: a **weekly cycle**
  - weekend RV is only 0.41 of weekday RV (Saturday 27% vs Wednesday 48% p.a.)
  - no equity or macro news, fewer institutional traders

MFM_ch9_acf

Implied vs Realised Volatility (1/2)



MFM_ch9_vix

Implied vs Realised Volatility (2/2)

- Left: VIX and SPY realised volatility over the next 21 trading days; right: the gap between them
- VIX: 30-day volatility implied by S&P 500 options (Chapter 12)
  - mean 19.52% vs 15.15% realised afterwards
  - VIX was higher on 88% of 1 468 days, by 4.37 percentage points on average
- **Variance risk premium** (VRP) Carr & Wu (2009), Bollerslev, Tauchen & Zhou (2009)
  - $\mathrm{VRP}_t = \mathbb{E}_t^{\mathbb{Q}}[\mathrm{RV}_{t,t+\mathbf{21}}] - \mathbb{E}_t^{\mathbb{P}}[\mathrm{RV}_{t,t+\mathbf{21}}]$: risk-neutral minus physical expectation
  - $\mathrm{RV}_{t,t+\mathbf{21}}$: realised variance over the next 21 trading days; $\mathrm{VIX}^{\mathbf{2}}$ approximates the first term
  - the chart shows an **ex-post** spread, a noisy proxy of this ex-ante premium
  - negative spread (March–April 2025): a forecast error, not a negative premium

MFM_ch9_vix

Implied Volatility as a Variance Forecast

- As a forecast: regress $\frac{252}{21}\mathrm{RV}_{t,t+21}$ (total variance, annualised %^{2}) on VIX_t^2 (annualised %^{2}; 30 calendar days $\approx$ 21 trading days)
  - slope 0.56 (standard error, SE, 0.05) < 1: VIX^2 moves more than the variance that follows, so VIX over-reacts (partly because the premium itself rises in turbulent periods); $R^2 = 0.23$: VIX is informative
  - level bias is a separate question, answered by the mean gap of 4.37 percentage points: VIX is on average above realised volatility (the premium); the same slope-versus-level distinction returns in the Mincer–Zarnowitz test

MFM_ch9_vix

The HAR-RV Model

- Corsi (2009): $\mathrm{RV}_{t+\mathbf{1}} = \beta_{\mathbf{0}} + \beta_d \mathrm{RV}_t + \beta_w \mathrm{RV}_t^{(w)} + \beta_m \mathrm{RV}_t^{(m)} + \varepsilon_{t+\mathbf{1}}$
  - $\mathrm{RV}_t^{(w)} = \frac{\mathbf{1}}{\mathbf{5}} \sum_{j=\mathbf{0}}^{\mathbf{4}} \mathrm{RV}_{t-j}$ (week), $\mathrm{RV}_t^{(m)} = \frac{\mathbf{1}}{\mathbf{22}} \sum_{j=\mathbf{0}}^{\mathbf{21}} \mathrm{RV}_{t-j}$ (month)
  - $\beta_{\mathbf{0}}$: constant; β_d, β_w, β_m: weights of yesterday's RV and of last week's and last month's average RV; $\varepsilon_{t+\mathbf{1}}$: forecast error
  - Bitcoin trades every day: week = 7 days, month = 30 days
- Heterogeneous market hypothesis Müller et al. (1997) (the "H" of HAR; HAR-RV: HAR model of realised variance): daily, weekly and monthly traders; a few components with different persistence mimic long memory
- An AR(22) with only 3 free slopes: weight $\beta_d + \beta_w/5 + \beta_m/22$ on lag 1, $\beta_w/5 + \beta_m/22$ on lags 2–5, $\beta_m/22$ on lags 6–22
- Estimation by ordinary least squares (OLS); errors are heteroskedastic and autocorrelated: use Newey–West standard errors (HAC: heteroskedasticity- and autocorrelation-consistent)
  - log-HAR: the same regression for $\ln \mathrm{RV}$, with the weekly and monthly terms as averages of $\ln \mathrm{RV}$; forecast $\exp(\hat{y} + \hat{s}^{\mathbf{2}}/2)$, $\hat{y}$ the fitted $\ln \mathrm{RV}$, $\hat{s}^{\mathbf{2}}$ the residual variance: the conditional mean if the errors are $N(0, s^{\mathbf{2}})$ (lognormal approximation)
  - log-HAR averages are geometric means of variances, not arithmetic means
  - For non-Normal errors use the smearing factor $\frac{\mathbf{1}}{T} \sum_t e^{\hat{\varepsilon}_t}$ Duan (1983); with heteroskedastic errors both are approximations

HAR Estimates: SPY and Bitcoin

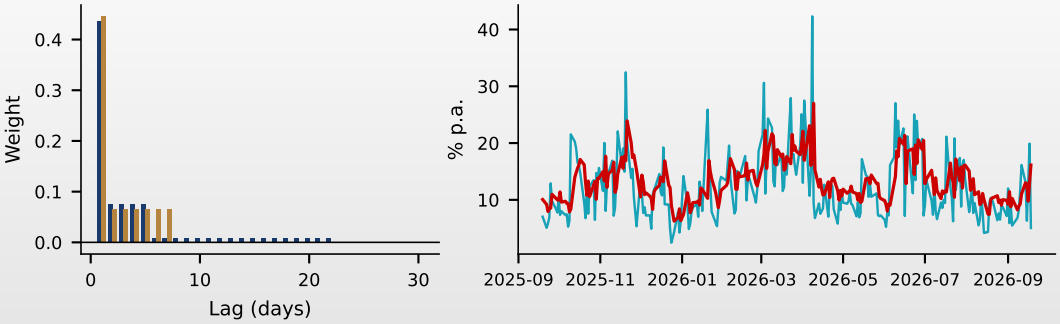
	β_0	β_d	β_w	β_m	R^2
SPY, HAR (levels)	0.273	0.399	0.222	0.114	0.31
	(0.064)	(0.089)	(0.096)	(0.053)	
SPY, log-HAR	−0.079	0.362	0.337	0.167	0.47
	(0.027)	(0.037)	(0.048)	(0.044)	
Bitcoin, HAR (levels)	1.496	0.412	0.409	−0.065	0.34
	(0.412)	(0.089)	(0.122)	(0.097)	
Bitcoin, log-HAR	0.223	0.380	0.465	−0.022	0.36
	(0.094)	(0.041)	(0.080)	(0.098)	

- Full samples; SPY: total daily variance (intraday RV plus squared overnight return), 1 467 days; Bitcoin: 24-hour RV, 547 days on which all three HAR regressors are available
- Newey–West standard errors in brackets

Reading the HAR Estimates

- SPY: all three components are significant; in logs $\hat{\beta}_d = 0.362$, $\hat{\beta}_w = 0.337$, $\hat{\beta}_m = 0.167$
  - Persistence $\hat{\beta}_d + \hat{\beta}_w + \hat{\beta}_m = 0.87$: shocks fade slowly, as in GARCH (Chapter 5)
- $R^2 = 0.47$ for log variance vs 0.31 for variance in levels: not comparable, the dependent variables differ; the out-of-sample losses on a common target (section Forecasting: HAR vs GARCH) decide
- Bitcoin: daily and weekly components matter; the monthly coefficient is close to 0 (-0.022, SE 0.098)
  - Short sample (two years) and a strong weekly cycle: the week absorbs most of the memory

Implied Lag Weights and In-Sample Fit (1/2)



MFM_ch9_har

Implied Lag Weights and In-Sample Fit (2/2)

- Left: weights of log-HAR on each past day (Bitcoin: complete 7- and 30-day windows); right: SPY, last 250 days, realised volatility and log-HAR fitted values
- SPY: lag 1 gets $\hat{\beta}_d + \hat{\beta}_w/5 + \hat{\beta}_m/22 = 0.362 + 0.067 + 0.008 = 0.437$; each of lags 2–5 gets $0.067 + 0.008 = 0.075$; each of lags 6–22 gets 0.008
  - The step-shaped weights decay slowly: the monthly component adds $\hat{\beta}_m/22 = 0.0076$ to each of lags 1–22, so $22 \times 0.0076 \approx 0.167 = \hat{\beta}_m$; days 6–22 alone carry $17 \times 0.0076 = 0.129$
  - Three coefficients mimic the slow decay of the autocorrelation function
- Bitcoin: yesterday 0.446, days 2–7 about 0.066 each, almost nothing beyond a week
- In-sample fitted values follow the level of volatility, not the day-to-day noise
- HAR is an AR(22) with 19 restrictions ($\phi_2 = \ldots = \phi_5$, $\phi_6 = \ldots = \phi_{22}$); Newey–West Wald test on SPY: $W = 18.1$, $p = 0.51$, not rejected (derivation in the Appendix)

MFM_ch9_har

Extensions of HAR

- **HAR-CJ** Andersen, Bollerslev & Diebold (2007): separate continuous (C) and jump (J) parts as regressors
  - $\ln \mathrm{RV}_{t+1} = \beta_0 + \beta_{Cd} \ln C_t + \beta_{Cw} \ln C_t^{(w)} + \beta_{Cm} \ln C_t^{(m)}$
  - $\quad + \beta_{Jd} \ln(1 + J_t) + \beta_{Jw} \ln(1 + J_t^{(w)}) + \beta_{Jm} \ln(1 + J_t^{(m)}) + \varepsilon_{t+1}$
  - $C_t^{(w)}, J_t^{(w)}$ ($C_t^{(m)}, J_t^{(m)}$): arithmetic means over the last 5 (22) days, as $\mathrm{RV}_t^{(w)}$; logs are taken after averaging
- **HARQ** (HAR with realised quarticity, RQ) Bollerslev, Patton & Quaedvlieg (2016): less weight on yesterday's RV when it is imprecise
  - measurement error in RV_t attenuates the daily slope; imprecise = high realised quarticity (Appendix)
  - β_d becomes $\beta_d + \beta_Q \sqrt{\mathrm{RQ}_t}$ with $\beta_Q < 0$
- Weekend or holiday dummies, leverage (negative returns raise future RV), implied volatility as an extra regressor
- Multi-step: regress $\sum_{j=1}^{h} \mathrm{RV}_{t+j}$ directly on the same regressors

Which Model Forecasts Tomorrow's Variance Best?

- Target: tomorrow's daily variance; SPY: total variance, 3 January 2022 to 18 September 2026 (1 180 days); Bitcoin: 1 March 2025 onwards (422 days)
  - Every forecast uses only data available the evening before
- Five models
  - log-HAR and HAR on realised variance, re-estimated daily on an expanding window
  - GARCH(1,1) with Student-t errors on daily returns since 2000 (SPY) or 2014 (Bitcoin), re-estimated every 21 days (GARCH: Bollerslev (1986); Student-t errors: Bollerslev (1987))
  - EWMA (exponentially weighted moving average) with decay factor $\lambda = 0.94$ (Chapter 5) and "yesterday's RV" (random walk)
- The model confidence set adds SHAR (realised semivariances) and HARQ (slide Extensions of HAR)
- HAR sees intraday data; GARCH and EWMA see only daily returns

Evaluating Against a Noisy Target

- True variance is unobserved; we compare forecasts F_t with the proxy RV_t
  - For SPY the proxy is total variance $\mathrm{RV}_t + r^2_{\text{overnight},t}$
  - A loss is **robust** if the ranking by *expected* loss with a proxy that is conditionally unbiased (given the forecasts' information) equals the ranking with true variance Patton (2011)
- Two robust losses
  - MSE $= (\mathrm{RV}_t - F_t)^2$: dominated by the most volatile days
  - QLIKE $= \dfrac{\mathrm{RV}_t}{F_t} - \ln \dfrac{\mathrm{RV}_t}{F_t} - 1 \geq 0$: depends only on the ratio; penalises under-prediction more
- Common mistake: MAE (mean absolute error), or MSE on volatility instead of variance, is **not** robust
  - Andersen & Bollerslev (1998): against r_t^2, GARCH forecasts explained a few % of the variation; against RV the same forecasts explain far more: the proxy was noisy, not the model

Comparing Two Forecasts: Diebold–Mariano

- Diebold & Mariano (1995) (DM) test: loss difference $d_t = L(\mathrm{RV}_t, F_t^A) - L(\mathrm{RV}_t, F_t^B)$; $t = \bar{d}/\mathrm{se}_{\mathrm{NW}}(\bar{d}) \to N(0,1)$
  - $L(\mathrm{RV}_t, F_t)$: the loss (QLIKE or MSE) of forecast F_t; $\bar{d}$: mean of d_t; $\mathrm{se}_{\mathrm{NW}}$: Newey–West standard error
  - $t < 0$: model A has the lower loss; HAC standard error because d_t is autocorrelated (Chapter 8)
- What drives t: the mean gap **and** how stable the gap is from day to day
  - Check, ignoring autocorrelation, $T = 100$ days: a gap with mean 0.01 and standard deviation 0.02 gives $t = 0.01/(0.02/\sqrt{100}) = 5$; a gap with mean 0.03 and standard deviation 0.30 gives $t = 1$
  - two forecasts that move together have a stable gap: a small mean difference can then be highly significant

Calibration: Mincer–Zarnowitz

- Mincer & Zarnowitz (1969) (MZ): $\mathrm{RV}_t = a + b\,F_t + u_t$: regress the realised target on the forecast (a: intercept, b: slope, u_t: error); an unbiased forecast has $a = 0$, $b = 1$; $b < 1$: forecasts too extreme
  - $b > 1$: forecasts too timid: realised variance moves more across days than the forecast does
  - Check: a forecast of 2 on calm days and 4 on busy days, with realised means 1.5 and 4.5, has $b = (4.5 - 1.5)/(4 - 2) = 1.5 > 1$ and $a = 1.5 - 1.5 \times 2 = -1.5$: it under-separates the two kinds of day
  - Joint Wald test with Newey–West covariance; R^2 measures how much of the variation the forecast explains
  - R^2 is capped by the noise in RV: even the true variance would not reach 1 Andersen & Bollerslev (1998)

Results: SPY, 2022–2026

Model	QLIKE	DM_{QLIKE} t (p)	MSE ($\%^4$)	DM_{MSE} p	MZ b	MZ R^2
log-HAR	0.310	–	4.15	–	1.16	0.30
HAR	0.326	-1.95 (0.052)	4.48	0.117	0.75	0.26
GARCH(1,1)-t	0.343	-2.30 (0.021)	4.87	0.188	0.70	0.20
EWMA	0.393	-4.02 (< 0.001)	5.14	0.050	0.74	0.14
Yesterday's RV	0.502	-5.95 (< 0.001)	5.41	0.060	0.54	0.29

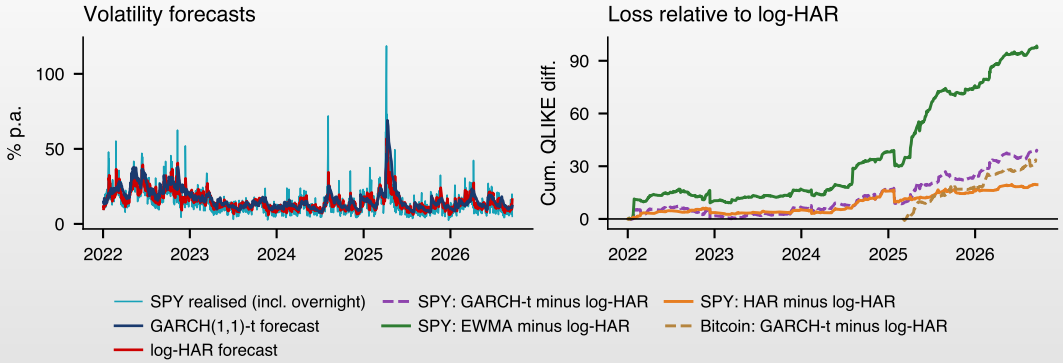
- 1 180 one-day-ahead forecasts of total daily variance; DM against log-HAR: $d_t = L(\text{log-HAR}) - L(\text{model})$ with $L = \text{QLIKE}$ or MSE, Newey–West; $t < 0$ favours log-HAR
- log-HAR has the lowest QLIKE and MSE: QLIKE 10% below GARCH(1,1)-t (DM $t = -2.30$, $p = 0.021$)
  - Against EWMA and yesterday's RV the gain is highly significant; against HAR in levels only marginally ($p = 0.052$)
- MSE differences are not significant at 5% (DM on MSE: p from 0.050 for EWMA, borderline, to 0.188 for GARCH): a few crisis days (April 2025) dominate the squared errors
- Mincer–Zarnowitz: log-HAR $b = 1.16$ ($p = 0.554$ for $a = 0, b = 1$), GARCH $b = 0.70$: no significant bias, but log-HAR explains more (R^2 0.30 vs 0.20)
  - log-HAR $b > 1$: if anything slightly too timid, but only 0.66 standard errors from 1 (SE 0.25); GARCH $b < 1$: slightly too extreme, also not significant
- Yesterday's RV has a high R^2 but $b = 0.54$: right direction, far too extreme

Results: Bitcoin, 2025–2026

Model	QLIKE	$\mathrm{DM}_{\mathrm{QLIKE}}$ t (p)	MSE (%^{4})	$\mathrm{DM}_{\mathrm{MSE}}$ p	MZ b	MZ R^2
log-HAR	0.282	–	26.6	–	0.82	0.38
HAR	0.409	-1.22 (0.223)	27.6	0.431	0.89	0.34
GARCH(1,1)-t	0.360	-4.32 (< 0.001)	42.2	0.055	0.55	0.23
EWMA	0.346	-2.96 (0.003)	36.3	0.033	0.64	0.18
Yesterday's RV	0.466	-5.38 (< 0.001)	35.0	0.061	0.58	0.33

- 422 one-day-ahead forecasts of 24-hour realised variance; same models and tests as for SPY
- log-HAR wins again: QLIKE 22% below GARCH(1,1)-t (DM $t = -4.32$, $p < 0.001$)
  - GARCH on daily data is biased: MZ $b = 0.55$ ($p < 0.001$): it overreacts to large daily moves
- HAR in levels is worse than GARCH under QLIKE (0.409 vs 0.360): after calm weekends it predicts too low a variance, and QLIKE punishes under-prediction
- Under MSE only EWMA is significantly worse than log-HAR at 5% ($p = 0.033$; GARCH $p = 0.055$)
- MSE is in $(\%^2)^2 = \%^4$ and grows with the square of the variance level, so Bitcoin and SPY MSE are not comparable; QLIKE depends only on RV_t/F_t and is comparable across markets

Forecasts over Time (1/2)



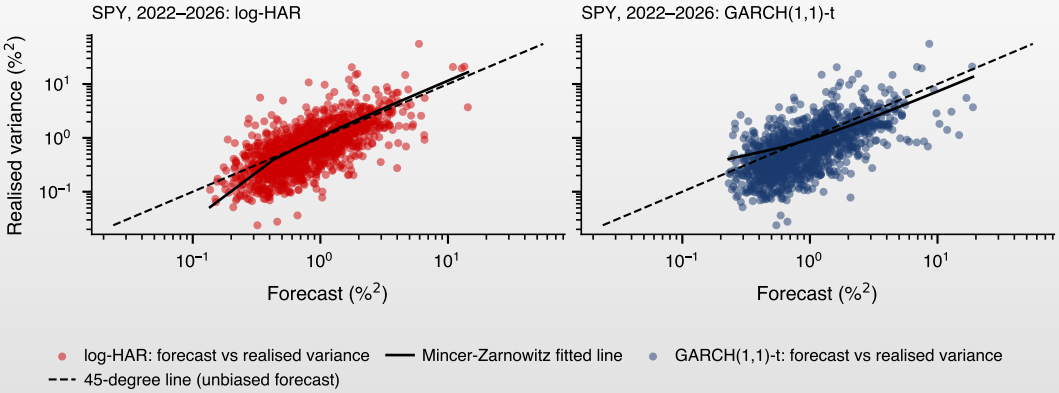
MFM_ch9_forecasts

Forecasts over Time (2/2)

- Left: SPY realised volatility (with overnight) and the two forecasts, annualised
- Right: cumulative QLIKE loss of each model minus log-HAR (rising = log-HAR better)
- After a shock both forecasts jump; GARCH stays high longer (its persistence is close to 1), log-HAR returns faster
  - April 2025: GARCH above 40% p.a. until late April
  - realised volatility below 20% from mid-April
- The cumulative loss difference rises steadily, with jumps at turbulent episodes: the gain is not driven by one day
- Of the three models plotted against log-HAR, EWMA loses most: a fixed $\lambda = 0.94$ reacts too much to single days

MFM_ch9_forecasts

Mincer–Zarnowitz Regressions (1/2)



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Mincer–Zarnowitz Regressions (2/2)

- SPY, 2022–2026: realised variance against the forecast (log scales); dashed: 45-degree line; black: fitted regression in levels
- log-HAR forecasts span a wider range: they separate calm from turbulent days better
- GARCH forecasts rarely fall below $0.26\%^2$ per day (1% of days), while log-HAR goes down to $0.14\%^2$: on calm days GARCH overstates the variance
- The cloud is wide for both: one day's variance is hard to predict, and RV itself is measured with error

MFM_ch9_forecasts

Which Models Survive? The Model Confidence Set

Model	QLIKE	MCS p-value	Eliminated at step	In MCS (90%)
log-HAR	0.310	1.000	–	✓
SHAR	0.320	0.224	–	✓
HAR	0.326	0.014	3	
HARQ	0.344	0.095	4–5	
GARCH(1,1)-t	0.343	0.095	4–5	
EWMA	0.393	0.012	2	
Yesterday's RV	0.502	0.000	1	

- Model confidence set (MCS) Hansen, Lunde & Nason (2011): test equal predictive ability, drop the worst model, repeat
  - range statistic, block bootstrap
  - the surviving set contains the best model with asymptotic probability $\geq 90\%$
- SPY, 1 180 days from 2022: the 90% set is {log-HAR, SHAR}
  - SHAR splits yesterday's RV into positive and negative realised semivariances Patton & Sheppard (2015) (Appendix)
- A multiple-comparison procedure: many pairwise DM tests without size control would overstate the evidence (Chapter 8)

Reading the Model Confidence Set

- An MCS p-value is a running maximum: if a model is eliminated at step k, its MCS p-value is $\max_{j \le k} p_j$, with p_j the p-value of the equal-ability test at step j
  - so the MCS p-values rise along the elimination order, which the table reads off; equal values (GARCH, HARQ) mean the later model's own step p-value was not larger, so their relative order is not identified from the table
  - a model stays in the 90% set if its MCS p-value exceeds 0.10
- Why is HAR (QLIKE 0.326) dropped before GARCH (0.343)? Each step eliminates the model with the largest **standardised** loss gap to the other survivors, not the largest mean loss
  - SHAR differs from HAR only in its daily term, so their losses move together and the gap is very stable: DM SHAR vs HAR $t = -3.37$, $p < 0.001$
  - GARCH loses more on average, but its gaps are noisier: against log-HAR $t = -2.30$, smaller in size than the gap that separates HAR from SHAR (as in the check on the Diebold–Mariano slide)
  - the DM table against log-HAR cannot show this: it never pairs HAR with SHAR, whose very precise gap is the likely reason HAR goes before GARCH

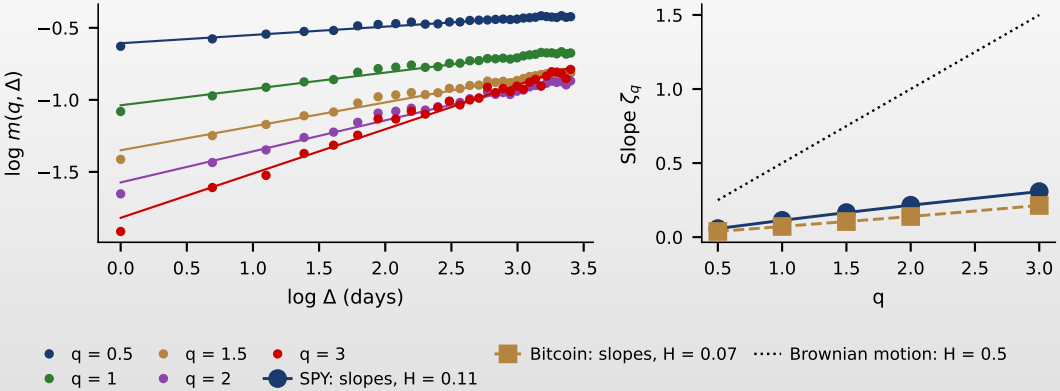
Is Volatility Rough?

- Gatheral, Jaisson & Rosenbaum (2018): log volatility behaves like a **fractional Brownian motion** with Hurst exponent $H \approx 0.1$
  - $H = 0.5$: ordinary Brownian motion; $H < 0.5$: increments are negatively correlated, paths are rougher; $H > 0.5$: smoother, trending
- Scaling law: $m(q, \Delta) = \frac{1}{T} \sum_t |\ln \sigma_{t+\Delta} - \ln \sigma_t|^q \propto \Delta^{qH}$
  - $m(q, \Delta)$: the average q-th power of the change in log volatility over Δ days; σ_t: daily volatility (here $\sqrt{\mathrm{RV}_t}$); $\propto$: proportional to
  - Regress $\ln m(q, \Delta)$ on $\ln \Delta$: slope $\zeta_q = qH$; then H from the slopes across q
- Rough models reproduce the shape of the implied volatility smile for short maturities (Chapter 12)

Fractional Brownian Motion and the Scaling Law

- Fractional Brownian motion B_H, $0 < H < 1$: a Gaussian process with $B_H(0) = 0$, mean 0 and $\mathrm{Cov}(B_H(t), B_H(s)) = \frac{1}{2}\left(|t|^{2H} + |s|^{2H} - |t-s|^{2H}\right)$
  - $H = 1/2$: the covariance is $\min(t, s)$, ordinary Brownian motion
  - increments: $B_H(t+\Delta) - B_H(t) \sim N(0, \Delta^{2H})$ for every t
- Model $\ln \sigma_t = \text{const} + \nu B_H(t)$: the change over Δ days is $\nu \Delta^H Z$ in distribution, $Z \sim N(0,1)$
  - $\mathbb{E}|\ln \sigma_{t+\Delta} - \ln \sigma_t|^q = \nu^q \, \mathbb{E}|Z|^q \, \Delta^{qH}$, so $\ln m(q, \Delta) \approx \ln(\nu^q \mathbb{E}|Z|^q) + qH \ln \Delta$: the log–log slope is $\zeta_q = qH$
- Estimate: slopes ζ_q for $q \in \{0.5, 1, 1.5, 2, 3\}$, then regress ζ_q on q through the origin: $\hat{H} = \sum_q q\,\zeta_q / \sum_q q^2$
- Check with $q = 2$: for Brownian motion $m(2, \Delta) = \nu^2 \Delta$, so doubling Δ doubles m and $\zeta_2 = 1 = 2 \times 0.5$; with $H = 0.1$, doubling Δ multiplies $m(2, \Delta)$ only by $2^{0.2} \approx 1.15$

Scaling of Log Volatility Increments (1/2)



MFM_ch9_rough

Scaling of Log Volatility Increments (2/2)

- Left: $\ln m(q, \Delta)$ against $\ln \Delta$ for SPY, $\Delta = 1, \ldots, 30$ days; right: slopes ζ_q against q
- The log–log points lie close to straight lines and the slopes are close to linear in q: consistent with the scaling law, which the plot alone cannot establish
  - $\hat{H} = 0.11$ for SPY (block bootstrap 95% CI $[0.08; 0.13]$), 0.07 for Bitcoin (only observed pairs at each calendar lag; interpolating the missing days gives 0.08): far below 0.5
- The catch: RV measures volatility with error; the next slides show how measurement error lowers $\hat{H}$

MFM_ch9_rough

Rough or Noisy? The Identification Problem

- The estimate $H \approx 0.1$ comes from RV, a noisy proxy of the latent σ_t
  - Cont & Das (2024): the scaling estimator applied to realised volatility finds roughness even when spot volatility is smooth: an estimation artefact
  - Bolko, Christensen, Pakkanen & Veliyev (2023) (generalised method of moments, GMM, that models the RV error) and Fukasawa, Takabatake & Westphal (2022) (consistent under high-frequency asymptotics) still find $H < 0.5$
- Formal question: $\ln \mathrm{RV}_t^{1/2} = \ln \sigma_t + e_t$, with $\ln \sigma_t$ fractional of index H and e_t the measurement error of RV
  - H_0: $H = 1/2$ (classical diffusive volatility) against H_1: $H < 1/2$
  - Identification problem: a small H and a large $\mathrm{Var}(e_t)$ both flatten $m(2, \Delta)$ at short lags, because $m(2, \Delta) = c\,\Delta^{2H} + 2\,\mathrm{Var}(e_t)$ for i.i.d. error (next slide)
- Short-run roughness can coexist with long memory Bennedsen, Lunde & Pakkanen (2022)
- Stakes: the short-maturity skew and volatility forecasts depend on H (Chapter 12)

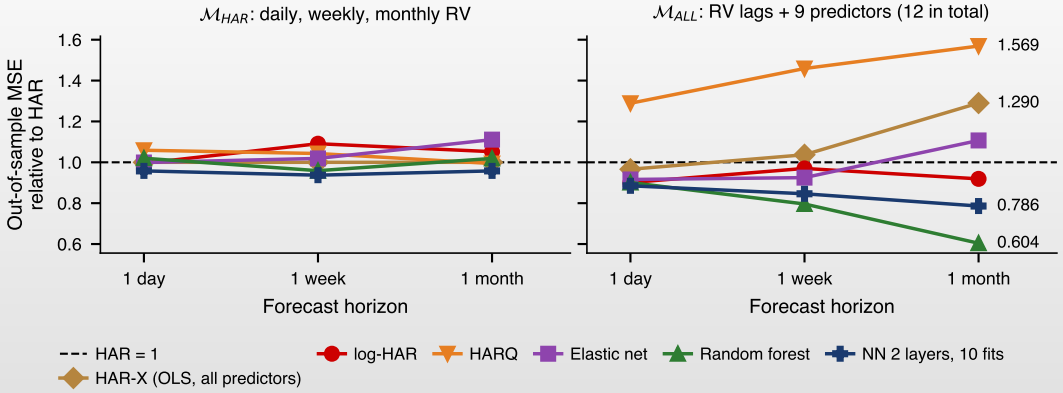
How Measurement Error Flattens the Scaling Law

- Observed increment = latent increment + error increment: $(\ln \sigma_{t+\Delta} - \ln \sigma_t) + (e_{t+\Delta} - e_t)$
  - if e_t is i.i.d. and independent of σ, the variances add: $\mathrm{Var}(e_{t+\Delta} - e_t) = 2\,\mathrm{Var}(e_t)$ for every $\Delta \geq 1$
  - hence $m(2, \Delta) = c\,\Delta^{2H} + k$, $c = \nu^2$, $k = 2\,\mathrm{Var}(e_t)$: a constant that does not grow with Δ
- Toy: $H = 0.5$, $c = 1$, $k = 1$
  - $m(2,1) = 1 + 1 = 2$, $m(2,2) = 2 + 1 = 3$: log–log slope $\ln(3/2)/\ln 2 = 0.58$, so $\hat{H} = 0.58/2 = 0.29$ instead of 0.5
  - at long lags the constant matters less: from $\Delta = 20$ to 40, slope $\ln(41/21)/\ln 2 = 0.97$, $\hat{H} = 0.48$
- Testable implication: under i.i.d. error, $\hat{H}$ rises when short lags are dropped; the mini-case in section AI for Scientific Discovery runs exactly this check on SPY

Case Study: Can Machine Learning Beat HAR?

- Christensen, Siggaard & Veliyev (2023): research questions
  - does off-the-shelf machine learning (ML) forecast daily realised variance better than the HAR family?
  - if so, why?
  - HAR Corsi (2009) is linear in three RV averages, while RV is right-skewed and persistent: a nonlinear map, or predictors beyond past RV, may help
- Data: 29 of the 30 stocks of the Dow Jones Industrial Average (DJIA), January 2001 – December 2017, $T = 4\,257$ days, 5-minute RV ($n = 78$)
  - Chronological split: training 70%, validation 10%, test 20%
- Information sets: $\mathcal{M}_{\text{HAR}}$ (daily, weekly, monthly RV) and $\mathcal{M}_{\text{ALL}}$, which adds 9 predictors: implied volatility, VIX, volume, momentum, macro and uncertainty indices
- Models, with minimal tuning: the HAR family by OLS, penalised regressions, tree ensembles (bagging, random forest, boosting) and neural networks; full catalogue in the Appendix
- Evaluation: out-of-sample MSE relative to HAR, per stock and then averaged; one-sided DM test per stock; MCS at 90%

Case Study: Gains by Horizon (1/2)



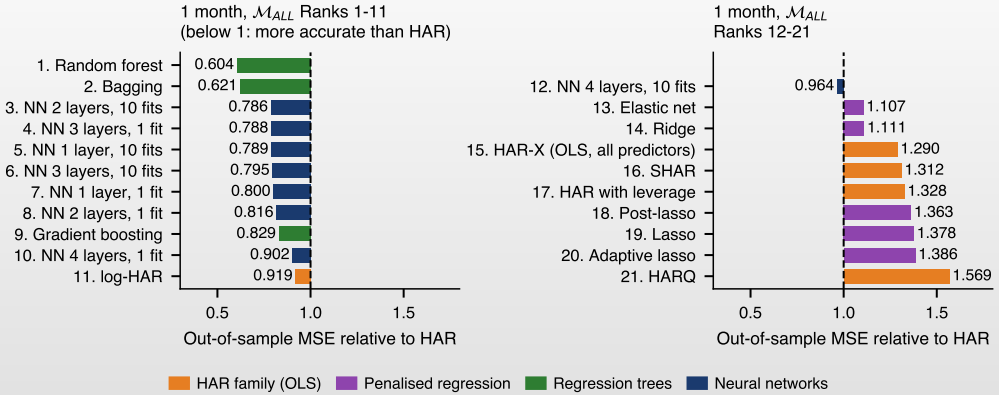
MFM_ch9_ml_vs_har

Case Study: Gains by Horizon (2/2)

- Source: Christensen, Siggaard & Veliyev (2023): out-of-sample MSE relative to HAR, averaged over 29 DJIA stocks (test set: the last 20% of 2001–2017); left: RV lags only, right: all 12 predictors; HAR-X: OLS on all predictors
- RV lags only: the six models shown stay between 0.937 and 1.111 times the HAR MSE at all three horizons: three RV averages leave little for a flexible model to learn
- All 12 predictors: the gain of trees and networks grows with the horizon
  - random forest 0.901 $\rightarrow$ 0.796 $\rightarrow$ 0.604
  - network with 2 hidden layers, averaged over 10 fits (NN_2^{10}): 0.885 $\rightarrow$ 0.846 $\rightarrow$ 0.786
  - the authors attribute it to the higher persistence of the ML fits
- Linear models with the same predictors deteriorate (HAR-X 0.966 $\rightarrow$ 1.290, HARQ 1.289 $\rightarrow$ 1.569)
  - Among the linear models, only log-HAR stays below 1 at all horizons (0.901, 0.970, 0.919)

MFM_ch9_ml_vs_har

Case Study: Ranking of 21 Models, One Month Ahead (1/2)



MFM_ch9_ml_vs_har

Case Study: Ranking of 21 Models, One Month Ahead (2/2)

- Source: Christensen, Siggaard & Veliyev (2023)
  - MSE relative to HAR, one month ahead, all 12 predictors, mean over 29 DJIA stocks
  - NN_k^e: k hidden layers, e networks averaged
- Tree ensembles lead: random forest 0.604, bagging 0.621
  - all eight networks beat HAR (0.786–0.964), but depth does not help
- Penalised regressions and linear HAR extensions lose to HAR
  - 1.107–1.569; the one linear exception, log-HAR, is on the horizon chart
- Each bar averages per-stock MSE ratios
  - not a significance test of the ranking

MFM_ch9_ml_vs_har

The Discovery Loop on the Roughness Question

- Question: is $\hat{H} \approx 0.1$ a property of σ_t or of the RV error e_t? H_0: $H = 1/2$ against H_1: $H < 1/2$
- One pass of the loop with an AI assistant (reusable prompts in the Appendix)
  - Literature: tabulate the published estimates of H (estimator, frequency, treatment of the RV error); check every DOI (digital object identifier) on Crossref
  - Competing hypotheses, each with its shape of $m(2, \Delta)$: fractional with $H = 0.1$; $H = 0.5$ plus i.i.d. error; two factors, one fast mean-reverting
  - Replicate the chapter's $\hat{H} = 0.11$ before extending; fix lag windows, measures and the fit with a constant noise term k in advance
- Checks the researcher owns
  - Proxy error: simulate $H = 0.5$ plus error of the same size; the procedure must not return $H \approx 0.1$
  - Overlapping increments invalidate OLS standard errors of ζ_q: use a block bootstrap; report every window tried
  - Assistants quote "$H \approx 0.1$ for all assets": check each number against the cited table

Mini-Case: Is $\hat{H}$ an Artefact of i.i.d. Error?

- A typical assistant proposal: $\hat{H} \approx 0.1$ comes from i.i.d. measurement error in RV. Implication: $m(2, \Delta) = c\,\Delta^{2\hat{H}} + k$ with $k = 2\,\mathrm{Var}(e_t) > 0$, so $\hat{H}$ **rises** when short lags are dropped

	$\Delta \in [1, 30]$	$[5, 30]$	$[10, 50]$	Fit with k, $[1, 50]$
Simulated $H = 0.5$ plus error	0.18	0.24	0.26	0.39
SPY, total RV	0.106	0.089	0.077	0.096
SPY, intraday RV	0.129	0.094	0.078	0.113

- First column replicates the chapter; the simulated row (one path, not a power study) shows the mechanism: $\hat{H}$ rises and the fit recovers most of the gap to 0.5
- SPY: $\hat{H}$ **falls** with the window and the fitted k is 0; $\hat{H} = 0.096$, block bootstrap 95% CI $[0.058; 0.268]$
- Interpretation: an illustrative diagnostic, inconsistent with the i.i.d.-error explanation; a formal rejection needs a simulation-calibrated test; the non-i.i.d. mechanism of Cont & Das (2024) is not tested by this design.

MFM_ch9_ai_discovery_case

Your Turn: A Project Seed

- Question: does the roughness of SPY and Bitcoin volatility survive an estimator that models the RV measurement error?
- Replicate first: $\hat{H} = 0.11$ (SPY) and 0.07 (Bitcoin) from the scaling law, and $\hat{H} = 0.18$ for simulated $H = 0.5$ plus error (Seminar 9, B9)
- Deliverables
  - A pre-registration (hypotheses, estimators, lag windows, tests, simulation design) dated before the first run
  - The GMM estimator of Bolko, Christensen, Pakkanen & Veliyev (2023) on the course RV series, with Monte Carlo size and power at $H \in \{0.1, 0.5\}$
  - Code, a two-page report with every specification tried, and a referee-style critique of your own result
- Log every prompt that shaped a design choice in `AI_USE.md` and every error the assistant made in `AI_ERRORS.md`: github.com/danpele/MFM/tree/main/project

Which Tool When?

Situation	Suggested tool
Liquid asset, 1–5 minute bars	RV at 5 minutes, subsampled
Tick data, or an illiquid stock (Bucharest Stock Exchange, BVB)	Pre-averaging or flat-top realised kernel ($n^{-1/4}$; Parzen $n^{-1/5}$; Appendix: rates of convergence); check the signature plot
Daily risk of an exchange-traded asset	Total variance: RV plus squared overnight return
Separate news jumps from diffusion	BV or MedRV (median realised variance; Appendix: jump-robust estimators), ratio test with FDR control; Lee–Mykland for the jump time
Daily or weekly variance forecast	log-HAR (plus jumps, weekends, RQ); benchmark GARCH
Comparing forecasts	QLIKE and MSE on variance, Diebold–Mariano, Mincer–Zarnowitz, model confidence set

- Strongest result: out of sample, log-HAR beats GARCH(1,1)-t under QLIKE by 10% (SPY) and 22% (Bitcoin); MSE comparisons are inconclusive
- The 5-minute rule is justified only if the signature plot is flat at 5 minutes and coarser
- The jump counts depend on FDR control and on the periodicity correction

Check Yourself, and Next: Chapter 10

Retrieval questions

- Why is R^2 of log RV not comparable with R^2 of RV in levels? (*Reading the HAR Estimates*)
- Why is QLIKE robust to a noisy RV proxy while MSE on volatility is not? (*Evaluating Against a Noisy Target*)
- Why does the MCS state a set of models rather than a winner? (*Which Models Survive?*)
- How can measurement error in RV make volatility look rough? (*How Measurement Error Flattens the Scaling Law*)

Chapter 10: market microstructure

- Where does the noise ε of this chapter come from: limit order books, bid–ask spreads, tick sizes
- Liquidity, market impact and high-frequency trading, on equity and crypto exchanges

Itô Semimartingales and Stable Convergence

- Itô semimartingale: $dp = \mu\, ds + \sigma\, dW + dJ$, with μ, σ adapted (known from the past at each time) and σ càdlàg (right-continuous with left limits, so it may jump); σ may depend on W (leverage)
  - Without jumps: $\sqrt{M}(\mathrm{RV}_t - \mathrm{IV}_t) \xrightarrow{\mathcal{L}\text{-}s} \mathrm{MN}(0, 2\,\mathrm{IQ}_t)$, a **mixed** Normal distribution whose variance $2\,\mathrm{IQ}_t$, with integrated quarticity IQ, $\mathrm{IQ}_t = \int \sigma^4(s)\, ds$, is itself random Barndorff-Nielsen & Shephard (2002)
- **Stable** convergence ($\mathcal{L}$-s): the limit holds jointly with every variable of the path's σ-field Mykland & Zhang (2009)
  - Hence $(\mathrm{RV}_t - \mathrm{IV}_t)/\sqrt{2\,\mathrm{RQ}_t/M} \to N(0,1)$: studentising by $\sqrt{2\,\mathrm{RQ}_t/M}$, with $\mathrm{RQ}_t \to \mathrm{IQ}_t$ random, is legitimate only under stable convergence
- Finite samples: the t-statistic for $\ln \mathrm{RV}_t$ has better coverage than the one in levels Barndorff-Nielsen & Shephard (2005); the confidence intervals of the lecture use it

Back

Deriving the Noise Bias and the Optimal Frequency

- Observed return $r_i = r_i^* + \varepsilon_i - \varepsilon_{i-1}$, with ε i.i.d. $(0, \omega^2)$ and independent of p^*
  - Given the path of σ (zero drift, no leverage: σ independent of W):
    $\mathbb{E}[\mathrm{RV}^{(n)} \mid \sigma] = \sum_i \mathbb{E}[r_i^{*2} \mid \sigma] + \sum_i \mathbb{E}[(\varepsilon_i - \varepsilon_{i-1})^2] = \mathrm{IV} + 2n\omega^2$; $\mathrm{cov}(r_i, r_{i-1}) = -\mathbb{E}[\varepsilon_{i-1}^2] = -\omega^2$
  - By a law of large numbers for the noise terms, $\mathrm{RV}^{(n)}/(2n) \xrightarrow{p} \omega^2$ as $n \to \infty$: the all-tick RV estimates the noise, not IV Bandi & Russell (2008)
- MSE: $\mathbb{E}[(\mathrm{RV}^{(n)} - \mathrm{IV})^2] \approx \underbrace{2\,\mathrm{IQ}/n}_{\text{discretisation}} + \underbrace{4n^2\omega^4}_{\text{squared bias}} + O(n)$
  - First-order condition $-2\,\mathrm{IQ}/n^2 + 8n\omega^4 = 0$ gives $n^* = (\mathrm{IQ}/(4\omega^4))^{1/3}$ Bandi & Russell (2008)

Back

Intraday Seasonality and the Jump Test

- Seasonality biases BV (not RV): $\mathbb{E}|r_i|\,|r_{i-\mathbf{1}}| \propto \sigma_i \sigma_{i-\mathbf{1}} \leq (\sigma_i^{\mathbf{2}} + \sigma_{i-\mathbf{1}}^{\mathbf{2}})/2$
  - Fast changes of scale (open, close) push $\mathrm{RV} - \mathrm{BV}$ up, so the ratio test (section Jumps) over-rejects
- Periodicity factor f_i, standard-deviation estimator of Boudt, Croux & Laurent (2011): scale each return by $\sqrt{\mathrm{BV}_t / M_t}$, take the root mean square by interval of the day, normalise the mean of $f_i^{\mathbf{2}}$ to 1
  - SPY: $f = 1.78$ at the open, 0.73 at the minimum, 1.52 at the close
- Dividing $r_{t,i}$ by f_i before the test (5% level: main text): at the 0.1% level 20 SPY rejections remain

Back

Rates of Convergence under Noise

Estimator	Error of order	Positive	Source
RV on all ticks	inconsistent (bias $2n\omega^2$)	yes	Bandi & Russell (2008)
Sparse RV, fixed 5-minute grid	$M^{-1/2}$, M fixed; bias stays	yes	Liu, Patton & Sheppard (2015)
Two-scale RV	$n^{-1/6}$	no	Zhang, Mykland & Aït-Sahalia (2005)
Multi-scale RV	$n^{-1/4}$	no	Zhang (2006)
Realised kernel, flat-top / Parzen	$n^{-1/4}$ / $n^{-1/5}$	no / yes	Barndorff-Nielsen, Hansen, Lunde & Shephard (2008)
Pre-averaging	$n^{-1/4}$	no	Jacod, Li, Mykland, Podolskij & Vetter (2009)

- $n^{-1/4}$ is the optimal rate with i.i.d. noise: the price of noise is half the usual exponent
- With $n = 23\,400$ seconds: $n^{1/4} \approx 12.4$, against $\sqrt{78} \approx 8.8$ for 5-minute RV: a heuristic only (rates ignore constants, noise size and the bias of sparse RV), consistent with the modest gains in Liu, Patton & Sheppard (2015)

Back

Pre-Averaging

- Local averages of $k_n \approx \theta\sqrt{n}$ returns (θ: a tuning constant, not the jump-test θ) with a weight g on $[0,1]$, e.g. $g(x) = \min(x, 1-x)$ Jacod, Li, Mykland, Podolskij & Vetter (2009)
  - $\bar{r}_i = \sum_{j=1}^{k_n - 1} g(j/k_n)\, r_{i+j}$: noise variance shrinks like $1/k_n$, the signal keeps order k_n/n
- Estimator, with $\psi_1 = \int_0^1 g'(x)^2 dx$, $\psi_2 = \int_0^1 g(x)^2 dx$ (1 and 1/12 for the triangular g; g': the derivative of g)
  - $\widehat{\mathrm{IV}} = \frac{1}{\theta\, \psi_2 \sqrt{n}} \sum_i \bar{r}_i^2 - \frac{\psi_1}{2\,\theta^2 \psi_2\, n} \sum_i r_i^2$ (second term: noise bias correction)
- Rate $n^{-1/4}$; the same device gives jump-robust (pre-averaged bipower) and covariance estimators
- Built for tick data: on 5-minute bars little noise is left to average out (empirical signature plot, section Microstructure Noise)

Back

Realised Covariance and Asynchronous Trading

- Realised covariance $\sum_i r^{(\mathbf{1})}_{t,i} r^{(\mathbf{2})}_{t,i} \to \int \sigma_{\mathbf{12}}(s)\, ds$ needs prices of both assets on the **same** grid; $r^{(\mathbf{1})}, r^{(\mathbf{2})}$: the returns of assets 1 and 2, $\sigma_{\mathbf{12}}(s)$: their spot covariance
  - Epps effect Epps (1979): measured correlation falls towards 0 as sampling gets finer, because trades are asynchronous and the last price is stale
- Hayashi & Yoshida (2005): sum the products of **all** pairs of returns whose time intervals overlap
  - $\sum_{i,j} r^{(\mathbf{1})}_i r^{(\mathbf{2})}_j \mathbb{1}\{(t_{i-\mathbf{1}}, t_i] \cap (s_{j-\mathbf{1}}, s_j] \neq \emptyset\}$: $(t_{i-\mathbf{1}}, t_i]$, $(s_{j-\mathbf{1}}, s_j]$: the intervals between consecutive trades of each asset; unbiased without synchronisation (in the absence of noise)
- With noise as well: pre-averaged or multivariate kernel versions; application: intraday portfolio risk (Chapter 4)

Back

Locating Jumps Within the Day, Lee–Mykland

- Lee & Mykland (2008): $L_{t,i} = r_{t,i}/\hat{\sigma}_{t,i}$, with $\hat{\sigma}^2_{t,i}$ the bipower average of the previous $K = 270$ returns (their value for 5-minute data)
  - Under no jumps, $\max_i |L_i|$ over n returns has a Gumbel limit: reject if $(|L| - C_n)/S_n > -\ln(-\ln(1-\alpha))$, with $C_n = \sqrt{2\ln n} - \frac{\ln\pi + \ln\ln n}{2\sqrt{2\ln n}}$ and $S_n = 1/\sqrt{2\ln n}$, the centring and scaling constants of the Gumbel limit (our $\hat{\sigma}^2$ includes the factor $\pi/2$, so the constant $c = \sqrt{2/\pi}$ of Lee & Mykland (2008) becomes 1)
  - SPY, returns divided by f_i, $\alpha = 1\%$, $n = 115\,453$ (the whole sample): threshold $|L| > 5.41$
- 169 jumps on 105 days; 62 of them on Wednesdays between 14:00 and 15:00, when Federal Open Market Committee (FOMC) statements (14:00) and press conferences (14:30) are released
  - Largest: $L = -22.9$ on 2025-10-29; only 10% of these days are also flagged by the daily ratio test
- The two tests ask different questions: one extreme return relative to local volatility, versus a daily gap between RV and BV

Back

Jump-Robust Estimators of Integrated Variance

- Truncated RV Mancini (2009): $\mathrm{TRV}_t = \sum_i r_{t,i}^{\mathbf{2}} 1\{|r_{t,i}| \le c\,\hat{\sigma}_t f_i\,\Delta^{\varpi}\}$, $\varpi \in (0, 1/2)$
  - Keep only the squared returns below a threshold: c: a multiple, $\hat{\sigma}_t$: a jump-robust daily volatility, f_i: the periodicity factor, ϖ: the exponent of the interval length Δ
  - Diffusive increments are $O_p(\Delta^{\mathbf{1/2}})$ (of order $\sqrt{\Delta}$) and end up below the threshold; jumps are $O(1)$ and are cut off
- MedRV Andersen, Dobrev & Schaumburg (2012): $\frac{\pi}{\mathbf{6-4\sqrt{3}+\pi}} \frac{M}{M-\mathbf{2}} \sum_i \mathrm{med}(|r_{i-\mathbf{1}}|, |r_i|, |r_{i+\mathbf{1}}|)^{\mathbf{2}}$
  - med: the median of three adjacent absolute returns; the first fraction makes MedRV consistent for IV
  - An isolated return larger than both neighbours never enters the median (which shifts to a neighbour), so a large isolated jump is excluded, though its effect is not removed exactly in finite samples; MedRV is less sensitive than BV to isolated zero returns; two adjacent jumps can still enter the median
- Jump share of SPY RV (RV minus the robust estimator), summed over all days without a test: BV 6.3%, MedRV 7.2%, TRV 12.5% (significant jump days only: 2.8%) ($c = 3$, $\varpi = 0.49$; truncation bites on 44% of days: c is a tuning choice)
- Aït-Sahalia & Jacod (2009): compare power variations $\sum |r|^p$, $p > 2$, at Δ and $k\Delta$; the ratio tends to 1 with jumps and to $k^{p/\mathbf{2}-\mathbf{1}}$ without: known limits under both hypotheses

Back

HAR as a Restricted AR(22), the Wald Test

- Unrestricted: $\ln \mathrm{RV}_{t+1} = \phi_0 + \sum_{k=1}^{22} \phi_k \ln \mathrm{RV}_{t+1-k} + \varepsilon_{t+1}$
  - HAR imposes $\phi_2 = \ldots = \phi_5$ (3 restrictions) and $\phi_6 = \ldots = \phi_{22}$ (16): $R\phi = 0$ with 19 rows
- Wald: $W = (R\hat{\phi})^\top (R\hat{V}_{\mathrm{NW}} R^\top)^{-1} R\hat{\phi} \to \chi^2_{19}$ under the HAR restrictions; R: the 19×23 restriction matrix, $\hat{V}_{\mathrm{NW}}$: Newey–West covariance of $\hat{\phi}$; large W = restrictions rejected
- SPY log total variance: $W = 18.1$, $p = 0.51$; R^2 0.477 (AR(22)) vs 0.470 (HAR)
  - The step restrictions are not rejected: HAR saves 19 parameters at almost no in-sample cost, and less estimation noise helps out of sample

Back

Measurement Error in the Regressors and HARQ

- $\mathrm{RV}_t = \mathrm{IV}_t + u_t$ with $\mathbb{E}[u_t \mid \sigma] = 0$ and $\mathrm{Var}(u_t \mid \sigma) = 2\,\mathrm{IQ}_t/M$ (the CLT of the section Realised Variance)
  - If $\mathrm{IV}_{t+\mathbf{1}} = \beta_{\mathbf{0}} + \beta\,\mathrm{IV}_t + \eta_{t+\mathbf{1}}$, OLS of $\mathrm{RV}_{t+\mathbf{1}}$ on RV_t gives plim $\hat{\beta} = \beta\,\dfrac{\mathrm{Var(IV)}}{\mathrm{Var(IV)} + \mathbb{E}[2\,\mathrm{IQ}/M]} < \beta$ for $\beta > 0$: attenuation ($\eta_{t+\mathbf{1}}$: the shock to IV; plim: probability limit)
- If the signal variance does not rise with IQ_t (a working approximation), the signal-to-noise ratio of RV_t falls when IQ_t is high; HARQ approximates the falling weight by $\beta_d + \beta_Q\sqrt{\mathrm{RQ}_t}$, $\beta_Q < 0$ Bollerslev, Patton & Quaedvlieg (2016)
- SPY, levels, Newey–West SE: intraday RV $\hat{\beta}_Q = -0.0088$ (0.0042); total variance -0.0022 (0.0024)
  - Full vector $(\hat{\beta}_{\mathbf{0}}, \hat{\beta}_d, \hat{\beta}_w, \hat{\beta}_m)$, SE in brackets; intraday RV: 0.015 (0.105), 1.135 (0.494), -0.120 (0.228), 0.026 (0.057); total variance: 0.217 (0.109), 0.542 (0.282), 0.173 (0.128), 0.088 (0.067)
  - Daily slope (intraday RV) 1.13 at the median $\sqrt{\mathrm{RQ}}$, 1.09 at its 99th percentile: the sign predicted by the theory, a small effect
  - Read $\hat{\beta}_d$ together with the other slopes: the daily slope above 1 is partly offset by a negative weekly slope, and the three RV averages overlap by construction (each contains yesterday's RV), so the split is imprecise (SE of $\hat{\beta}_d$: 0.494); the slopes sum to 1.04 (total variance: 0.80), so the intraday levels fit is an in-sample diagnostic for the sign of β_Q, not a forecasting model
  - Like for like, total variance: HAR $\hat{\beta}_d = 0.399$, HARQ $\hat{\beta}_d = 0.542$: letting the weight fall with RQ raises the base daily slope, in the direction the attenuation argument predicts (the difference is within one SE)

Back

Beyond HAR, Semivariances and Joint Return–RV Models

- Realised semivariances $\mathrm{RS}_t^{\pm} = \sum_i \boldsymbol{r}_{t,i}^{\boldsymbol{2}} 1\{r_{t,i} \gtrless 0\}$, $\mathrm{RS}^+ + \mathrm{RS}^- = \mathrm{RV}$; SHAR Patton & Sheppard (2015): $\beta_d \mathrm{RV}_t$ becomes $\beta^+ \mathrm{RS}_t^+ + \beta^- \mathrm{RS}_t^-$
  - SPY, levels (overnight return counted as one more return of the day): $\hat{\beta}^- = 0.56$ (0.23), $\hat{\beta}^+ = 0.21$ (0.18); Wald $\beta^+ = \beta^-$: $p = 0.35$
  - Bad volatility is more persistent, as in Patton & Sheppard (2015), but six years of data do not separate the two slopes
  - In the MCS comparison, HARQ and SHAR are estimated in levels with the filter of Bollerslev, Patton & Quaedvlieg (2016): forecasts outside the [min, max] range of the estimation-sample values are replaced by the sample mean
- Realized GARCH Hansen, Huang & Shek (2012): a GARCH equation driven by the realised measure x_t plus a **measurement equation**
  - $\ln h_t = \omega + \beta \ln h_{t-\boldsymbol{1}} + \gamma \ln x_{t-\boldsymbol{1}}$, $\ln x_t = \xi + \varphi \ln h_t + \tau(z_t) + u_t$, $r_t = \sqrt{h_t}\, z_t$: joint quasi-maximum likelihood
  - h_t: conditional variance; x_t: realised measure (e.g. RV); z_t: standardised shock; $\tau(z_t)$: leverage function; ξ, φ: measurement-equation coefficients, u_t: its error
- HEAVY (high-frequency-based volatility) models Shephard & Sheppard (2010): one GARCH-type equation for the daily variance, driven by lagged RV, and one for RV itself: consistent multi-step forecasts

Back

Case Study Data, 29 DJIA Stocks and 12 Predictors

- Data: 29 of the 30 DJIA stocks (Visa excluded), 29 January 2001 – 31 December 2017, $T = 4\,257$ days; 5-minute RV from cleaned New York Stock Exchange (NYSE) trade prices, $n = 78$ returns a day
  - Split: training 70% (2 964 days), validation 10% (424), test 20% (847); $2\,964 + 424 + 847 = 4\,235$: the first 22 days only feed the monthly RV average; robustness: training 1 000 or 2 000 days, validation 200
  - $\mathcal{M}_{\mathbf{HAR}}$: daily, weekly, monthly RV; $\mathcal{M}_{\mathbf{ALL}}$ adds 9 predictors: implied volatility, earnings-day dummy, 1-week momentum, change in log dollar volume; VIX, squared Hang Seng return, business-conditions index, change in the 3-month T-bill rate, policy-uncertainty index

Back

Case Study Models, from HAR to Neural Networks

- Models: HAR, log-HAR, HAR with leverage, SHAR, HARQ by OLS on a rolling window; ridge (L_2 penalty), lasso (L_1), elastic net (a mix), adaptive and post-lasso, penalties tuned on the validation set
  - Trees: bagging, random forest (RF) with Breiman's defaults, 500 trees, $J/3$ candidate predictors per split (J: number of predictors), minimum node size 5, gradient boosting
  - Neural networks NN_k^e: $k = 1, \ldots, 4$ hidden layers (2; 4-2; 8-4-2; 16-8-4-2 neurons), leaky ReLU, Adam, dropout, early stopping; $e = 1$: best of 100 seeds on validation, $e = 10$: average of the best 10
- Evaluation: out-of-sample MSE relative to HAR, averaged over stocks; one-sided DM test per stock Diebold & Mariano (1995); MCS at 90% Hansen, Lunde & Nason (2011)

Back

Case Study, Further Results

- One day ahead, RV lags only: MSE relative to HAR: log-HAR 0.995, HARQ 1.059, RF 1.020, neural networks 0.954–0.990
  - Past RV leaves little to learn: the networks give the clearest gains (1–5%), log-HAR barely beats HAR
- One day ahead, all 12 predictors: elastic net 0.916; the other models are on the horizon chart of the case study
  - Extra predictors pay most with shrinkage, nonlinearity or the log transform; HARQ overfits
- MCS at 90%: with all predictors, the networks stay in the set for about 80% of stocks, HAR for about 20–40%
  - Variable importance from accumulated local effects: daily RV, weekly RV and implied volatility dominate in all four models shown; RF and networks rank implied volatility higher than OLS with all predictors and elastic net

Back

Case Study, Replication, Limits and Extensions

- Replicate on course data: SPY 5-minute bars, open-to-close RV with 78 returns a day; the authors report that SPY confirms the stock results
  - The authors' robustness split: 1 000 training days, 200 validation days, the rest to 18 September 2026 as test set
  - Predictors: VIX (computed from S&P 500 index options; a proxy for SPY's own implied volatility, which would need SPY options), squared Hang Seng return, SPY momentum and dollar volume, T-bill and policy uncertainty (Federal Reserve Economic Data, FRED); no earnings dummy for a fund
- Limits stated by the authors
  - Minimal tuning: more tuning could improve ML further; the networks are trained once and not re-estimated in the test window
  - Rankings depend on the split: with 1 000 training days the financial crisis falls in the test set and the HAR extensions beat HAR
  - HAR models with a separate jump component are not included; accumulated local effects describe the average prediction and miss rare events such as earnings days
- Open extension: the paper ranks variance forecasts by MSE only
  - does the ML advantage survive QLIKE, which is robust to the noisy RV proxy Patton (2011)?
  - does it hold for Bitcoin's 24-hour RV?

Back

The Discovery Loop, Reusable Prompts (1/2)

- Tools: an AI assistant (a large language model, LLM, such as Claude, ChatGPT, Gemini or GitHub Copilot); literature search: Semantic Scholar, Elicit, Google Scholar
- **1. Literature**: tabulate the evidence, then check every DOI on Crossref (the title must match)
  - `Tabulate the estimates of H published since 2018: estimator, sampling frequency, treatment of the RV measurement error, point estimate and standard error, DOI.`
- **2. Competing hypotheses**, each with its implied shape of $m(2, \Delta)$
  - `Derive m(2, Delta) for three processes that give H near 0.1 with the scaling estimator: fractional log-volatility with H = 0.1; H = 0.5 plus i.i.d. error; a two-factor model with one fast mean-reverting factor.`
- **3. Data and code**: replicate the chapter's number first
  - `Write Python that builds daily total variance from the SPY 5-minute bars of the course data (5-minute RV plus the squared overnight return, previous close to today's open), computes m(q, Delta) for q in {0.5, 1, 1.5, 2, 3}, Delta = 1..30, and the slope H.` Target: $\hat{H} = 0.11$

Back

The Discovery Loop, Reusable Prompts (2/2)

- **4. Robustness**, pre-registered, with a power check
  - `Write a pre-registered plan: lag windows [1,30], [5,30], [10,50]; RV, bipower variation and 30-minute subsampled RV; a fit m(2, Delta) = c Delta^(2H) + k; block-bootstrap intervals; size and power by simulation at H = 0.1 and 0.5.`
- **5. Critique**: ask for the strongest counter-argument
  - `Act as a referee: which non-i.i.d. error structures in log RV (serially correlated, level-dependent, driven by overnight returns or jumps) would survive the constant-k correction, and how would you detect each?`
- **6. Report**: all windows and measures, bootstrap intervals, the simulation design, code and the AI log

Back

Further Reading

- Textbook and surveys
  - Franke, Härdle & Hafner (2019): statistics of financial markets, volatility models
  - Andersen, Bollerslev, Diebold & Labys (2003): modelling and forecasting realised volatility; Corsi (2009): the HAR model
- Crypto-assets
  - Intraday information can enter Bitcoin risk forecasts; the entropy of high-frequency returns has been proposed as one such signal Pele & Mazurencu-Marinescu-Pele (2019)
  - Crypto-assets form a distinct asset class by the features of their return distributions Pele et al. (2023)

Back

Further Study on Quantinar

Advanced courses on quantinar.com

- Nonparametric and Semiparametric Models
- Kalman Filter
- Applied Time Series Analysis with Python

Code and data

- Quantlets: github.com/danpele/MFM/tree/main/Quantlets/Ch_09

Acronyms used in this material (1/2)

- **ACF**: AutoCorrelation Function
- **AI**: Artificial Intelligence
- **AR**: AutoRegressive (model)
- **BV**: Bipower Variation
- **BVB**: Bursa de Valori București — *the Bucharest Stock Exchange*
- **CI**: Confidence Interval
- **CLT**: Central Limit Theorem
- **DJIA**: Dow Jones Industrial Average
- **DM**: Diebold–Mariano (test)
- **ETF**: Exchange-Traded Fund
- **EWMA**: Exponentially Weighted Moving Average
- **FDR**: False Discovery Rate
- **FOMC**: Federal Open Market Committee
- **FRED**: Federal Reserve Economic Data
- **FWER**: Family-Wise Error Rate
- **GARCH**: Generalised AutoRegressive Conditional Heteroskedasticity
- **GMM**: Generalized Method of Moments
- **HAC**: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- **HAR**: Heterogeneous AutoRegressive (model)
- **HAR-CJ**: HAR model with Continuous and Jump components
- **HAR-RV**: Heterogeneous AutoRegressive model of Realised Volatility

Acronyms used in this material (2/2)

- **HARQ**: HAR model with Realised Quarticity
- **HEAVY**: High-frEquency-bAsed VolatilitY (models)
- **IQ**: Integrated Quarticity
- **LLM**: Large Language Model
- **MAE**: Mean Absolute Error
- **MCS**: Model Confidence Set
- **ML**: Machine Learning
- **MSE**: Mean Squared Error
- **MZ**: Mincer–Zarnowitz (regression)
- **NYSE**: New York Stock Exchange
- **OLS**: Ordinary Least Squares
- **QLIKE**: Quasi-LIKElihood (loss function)
- **RF**: Random Forest
- **RK**: Realised Kernel
- **RQ**: Realised Quarticity
- **RV**: Realised Variance
- **SE**: Standard Error
- **SHAR**: Semivariance HAR model
- **TQ**: Tripower Quarticity
- **TRV**: Truncated Realised Variance
- **TSRV**: Two-Scale Realised Variance
- **US**: United States
- **UTC**: Coordinated Universal Time (Temps Universel Coordonné)
- **VIX**: CBOE Volatility IndeX
- **VRP**: Variance Risk Premium
- **HAR-X**: Heterogeneous AutoRegressive model with additional regressors
- **DOI**: Digital Object Identifier

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